

SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: Data Visualization with Power BI and Tableau

PRACTICAL NO. 11

AIM: Implementing Advanced Visualization Techniques and Customizations.

- a) Create slope charts and lollipop charts
- b) Design waterfall charts
- c) Build sparklines for trend analysis
- d) Implement sheet swapping for dynamic dashboards
- e) Create custom geographic territories
- f) Use background images in dashboards
- g) Apply animation and transparency techniques

A: Create Slope Charts and Lollipop Charts

1. Create a Slope Chart (Year-over-Year Progression)

Step 1: Set Up the Base Worksheet

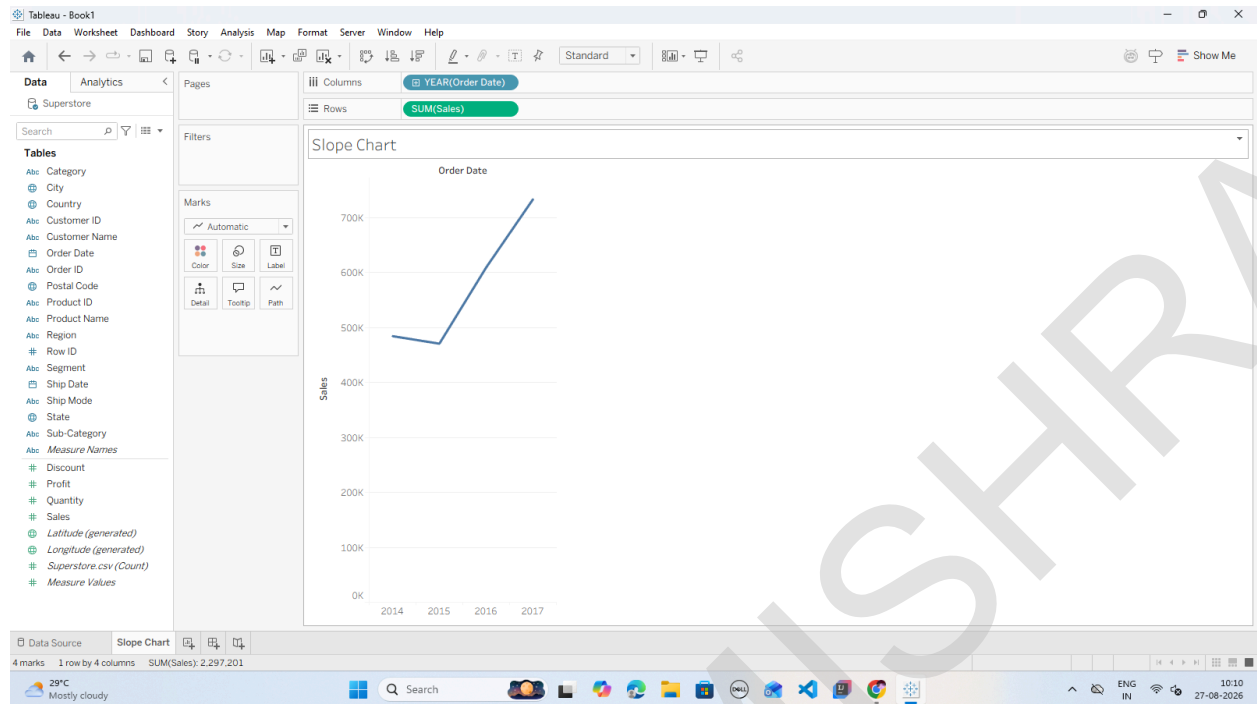
The screenshot displays the Tableau Desktop interface with the 'Superstore.csv' data source loaded. The interface is divided into several sections:

- Connections:** A sidebar on the left showing the 'Superstore' connection.
- Files:** A list of files including 'Department_Master.csv', 'housing.csv', 'Indian_Clima..._024_2025.csv', 'iris.csv', 'Movie Recommendations.csv', 'pima-indians..._etes.data.csv', 'Sem1_Scores.csv', 'Sem2_Scores.csv', 'startup_funding.csv', 'store_sales_data.csv', 'Student Socia...th Impact.csv', 'Student_Activities.csv', 'supermarket_sales.csv', 'Superstore.csv', and 'zomato.csv'.
- Data Source:** A section showing the 'Superstore.csv' data source with 21 fields and 9994 rows.
- Fields:** A table listing the fields in the data source, including 'Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode', 'Customer ID', 'Customer Name', and 'Segment'.
- Data Preview:** A table showing the first 11 rows of data, including columns for 'Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode', 'Customer ID', 'Customer Name', and 'Segment'.

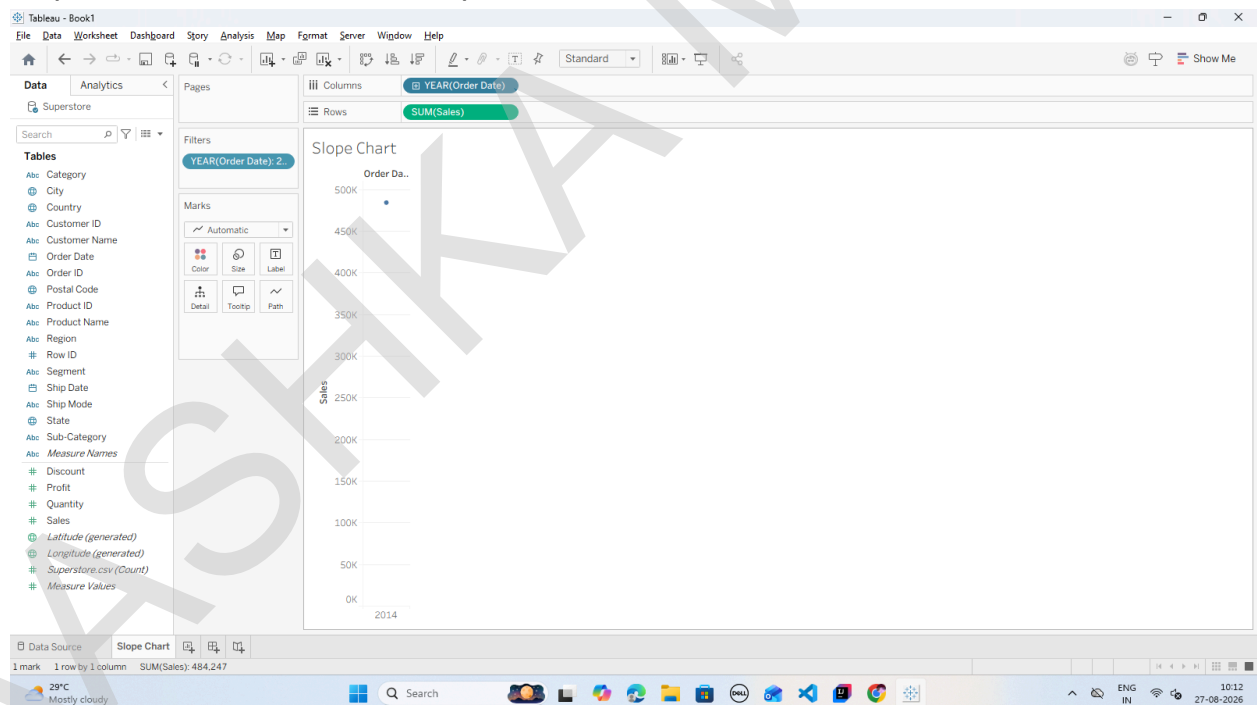
#	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment
1	CA-2016-152156	08-11-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consum	
2	CA-2016-152156	08-11-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consum	
3	CA-2016-138688	12-06-2016	16-06-2016	Second Class	DV-13045	Darrin Van Huff	Corporat	
4	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335	Sean O'Donnell	Consum	
5	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335	Sean O'Donnell	Consum	
6	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consum	
7	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consum	
8	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consum	
9	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consum	
10	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consum	
11	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consum	

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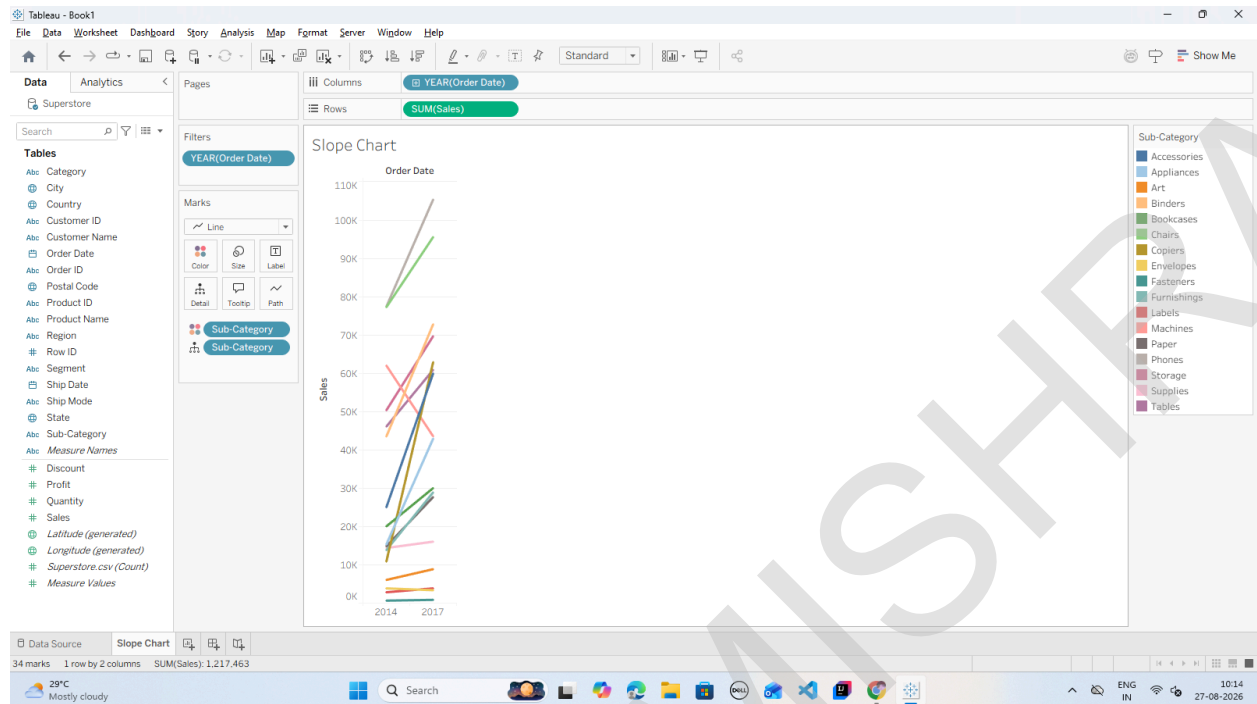
Step 2: Filter to Baseline & Comparison Years



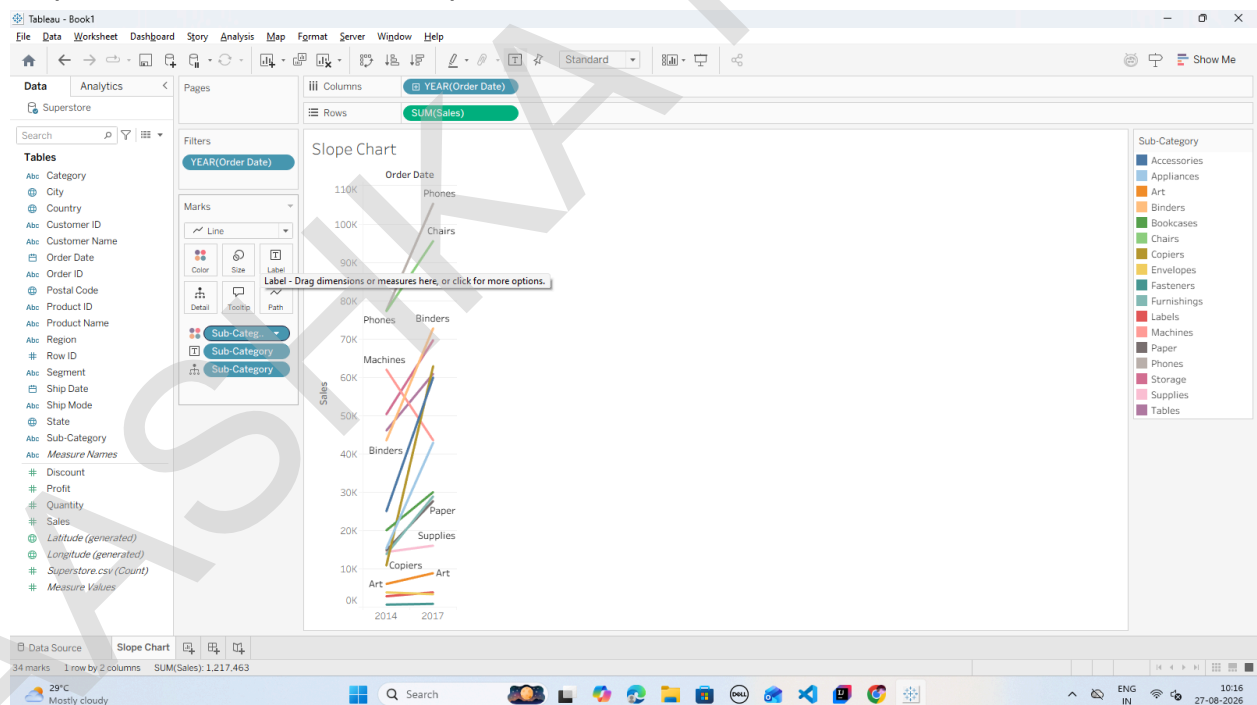
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Step 3: Disaggregate by Sub-Category



Step 4: Format Labels and Endpoints

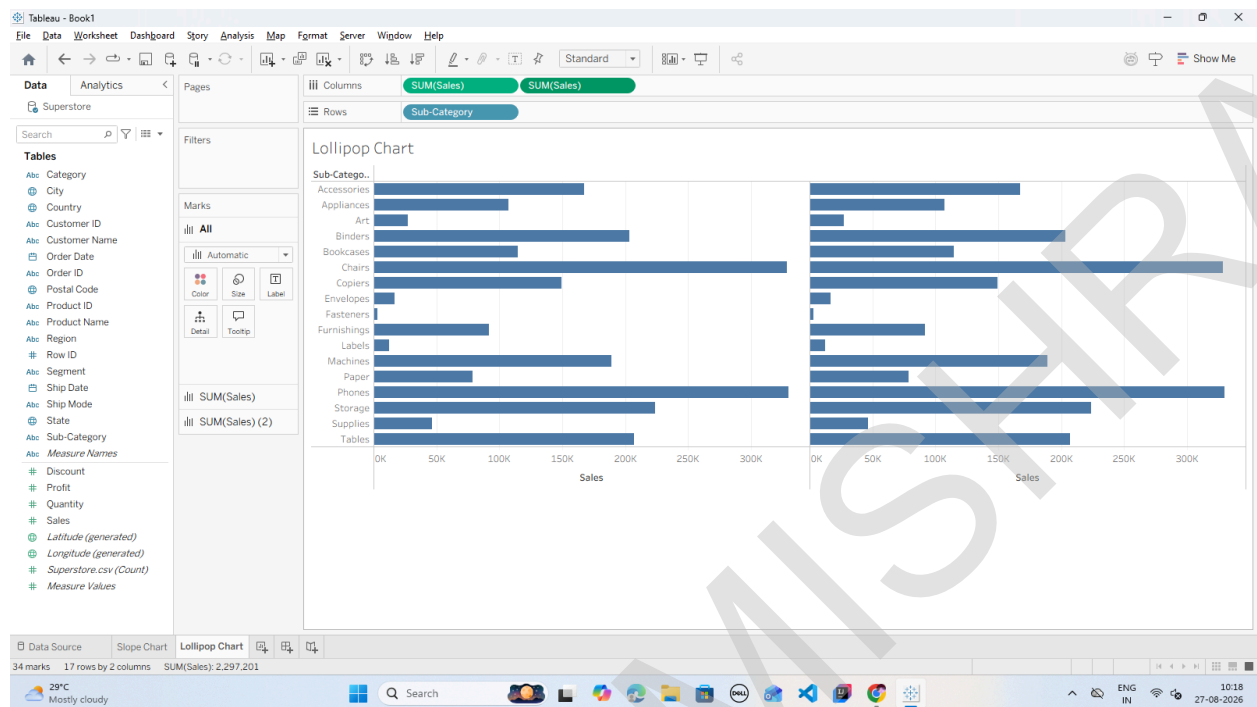


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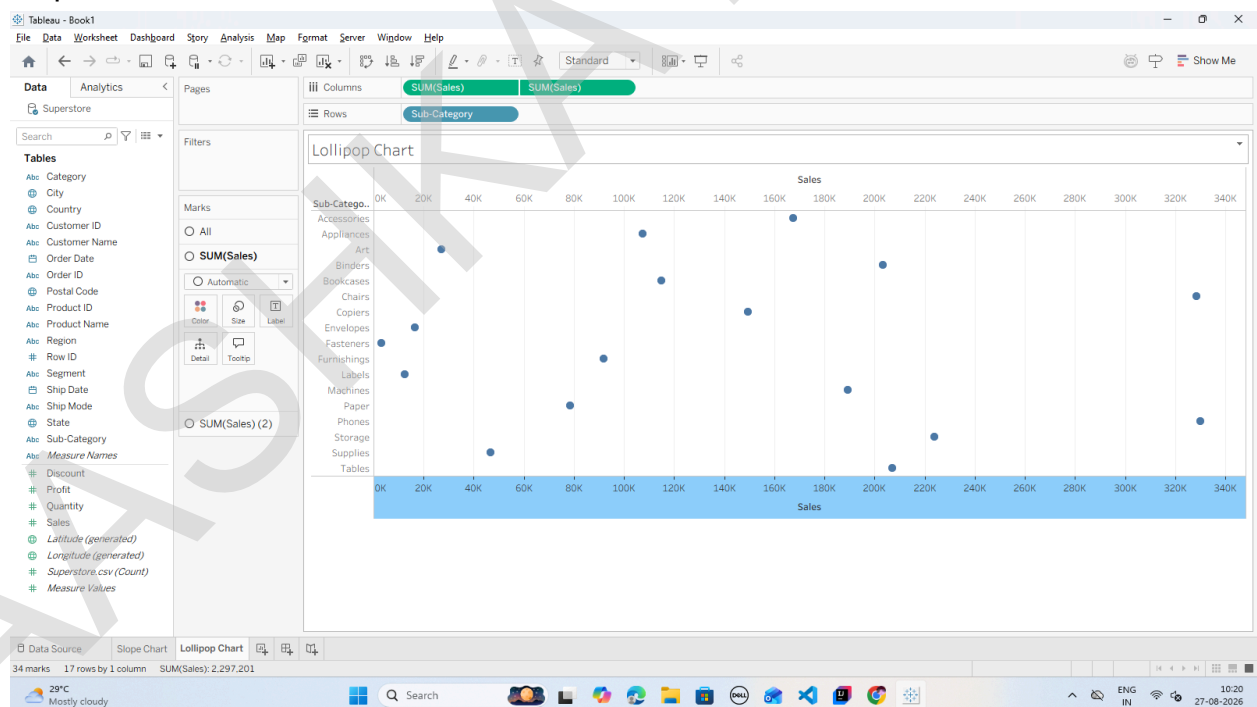
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2. Create a Lollipop Chart (Sales with Profit Highlighting)

Step 1: Set Up the Base Worksheet



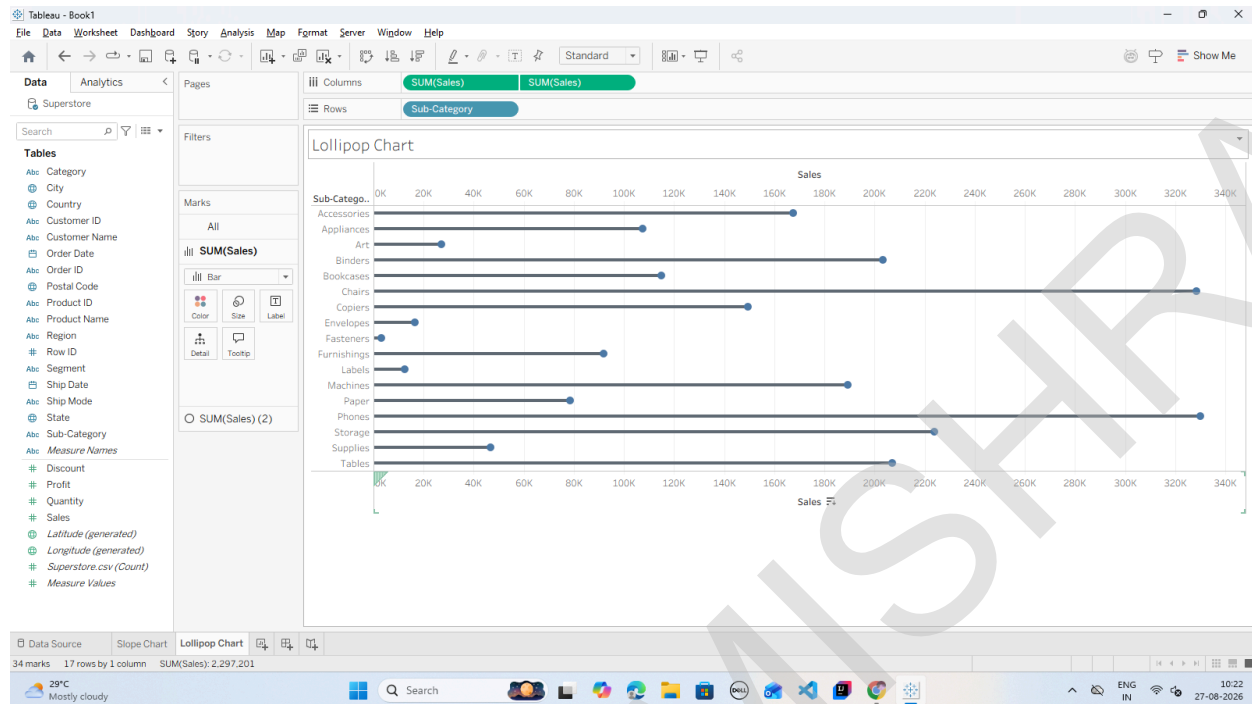
Step 2: Convert to Dual Axis



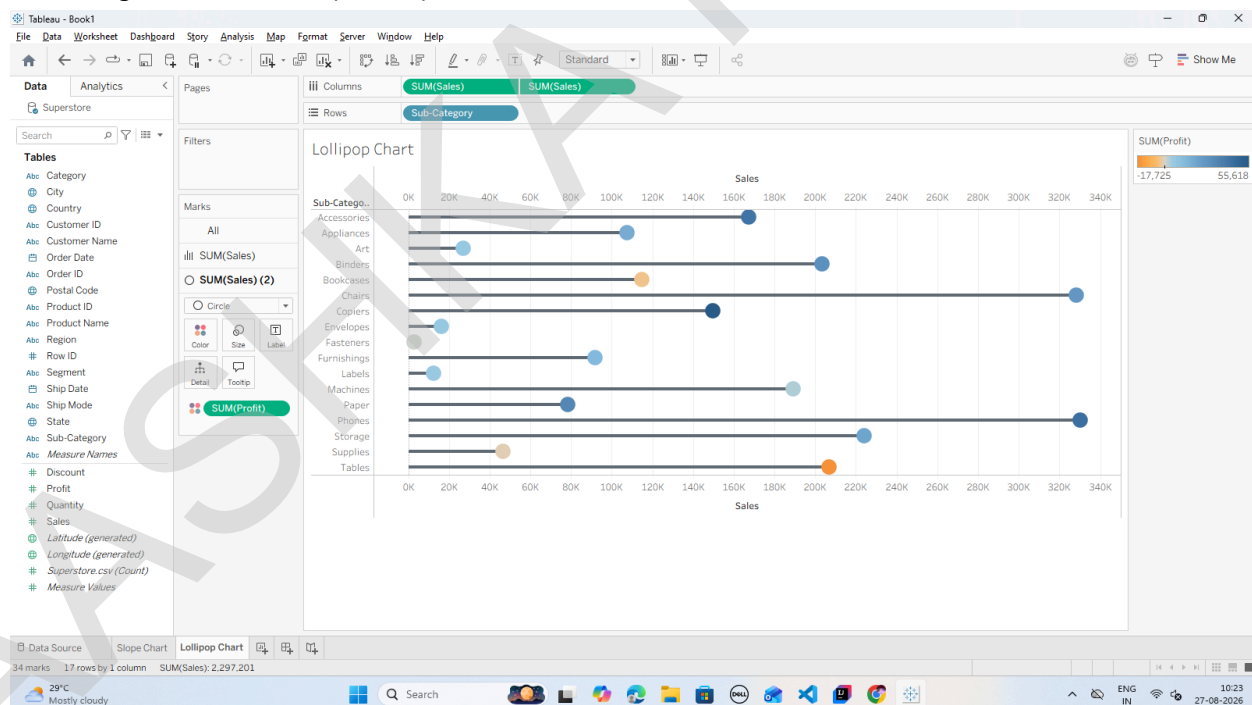
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Step 3: Configure the Marks Cards



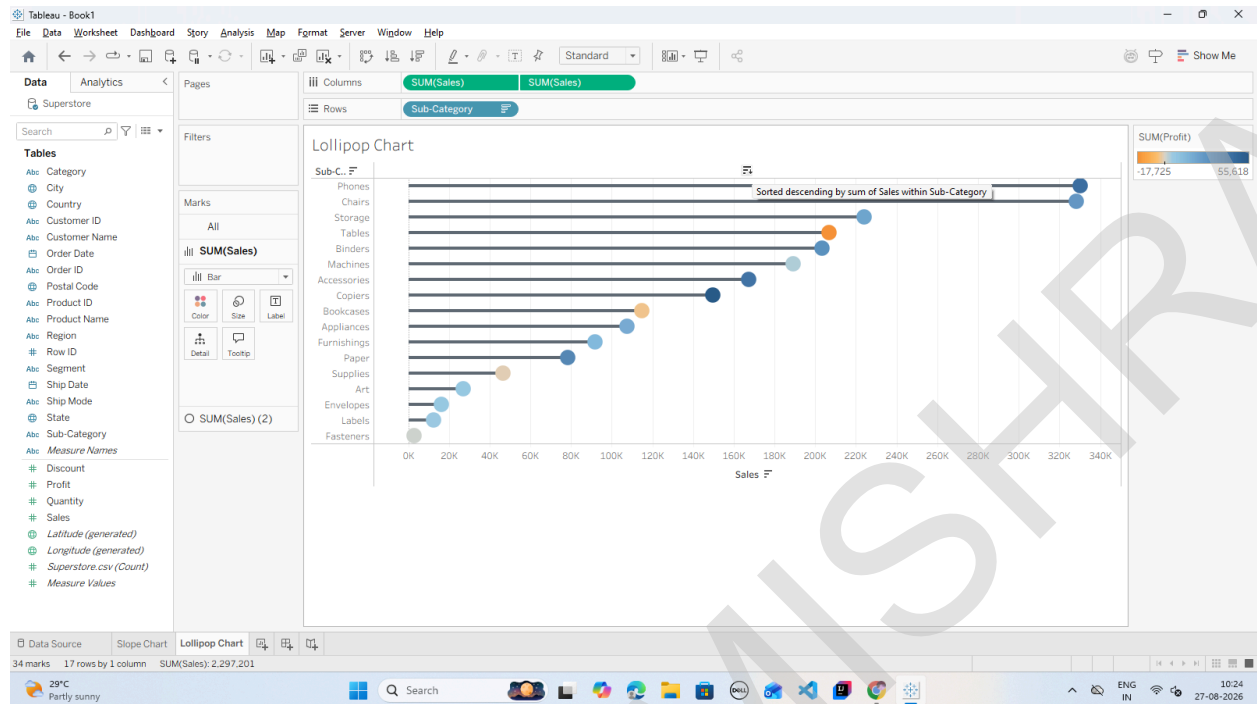
2. Configure the Head (Circle):



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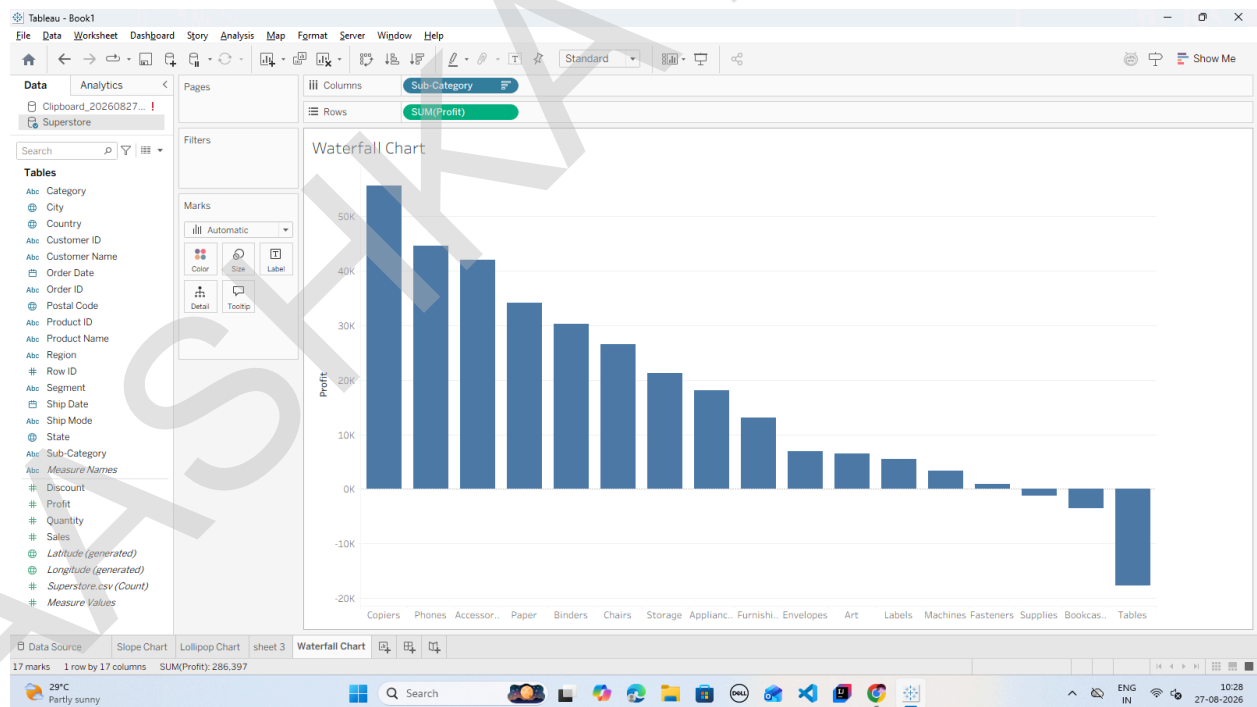
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Step 4: Clean Up Axis Formatting



B: Design Waterfall Charts

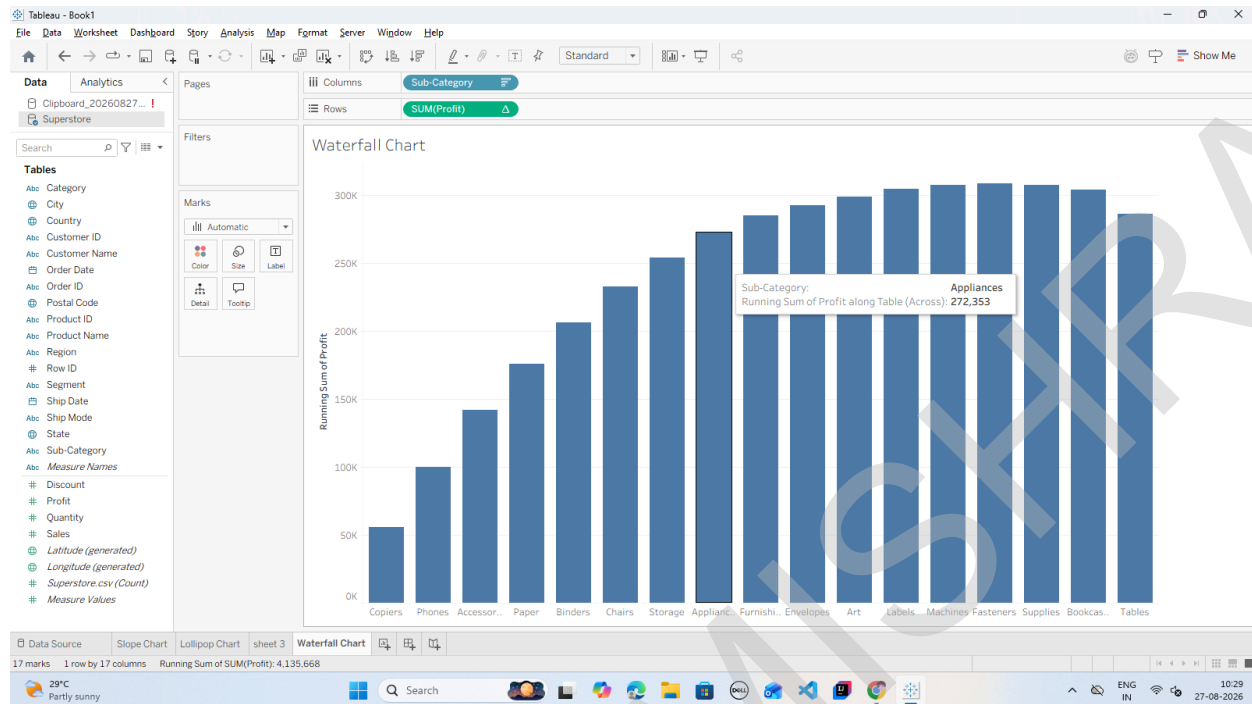
Step 1: Set Up the Base Worksheet



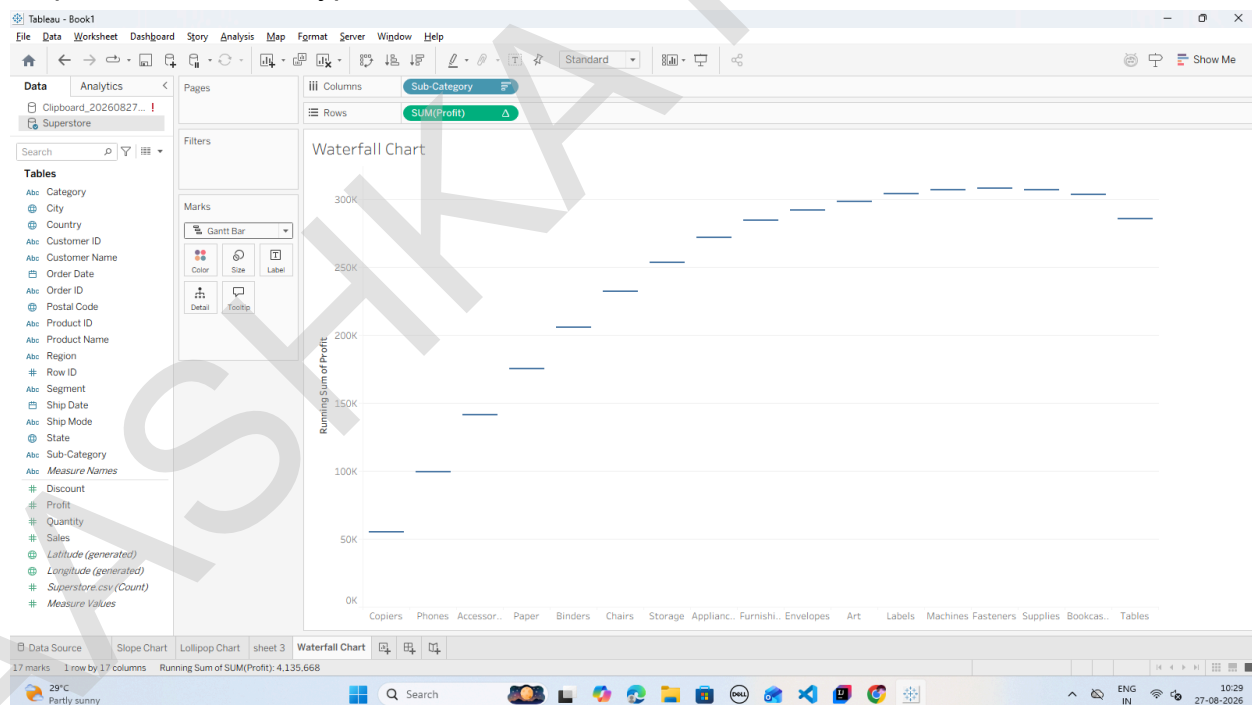
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Step 2: Apply Running Total Table Calculation



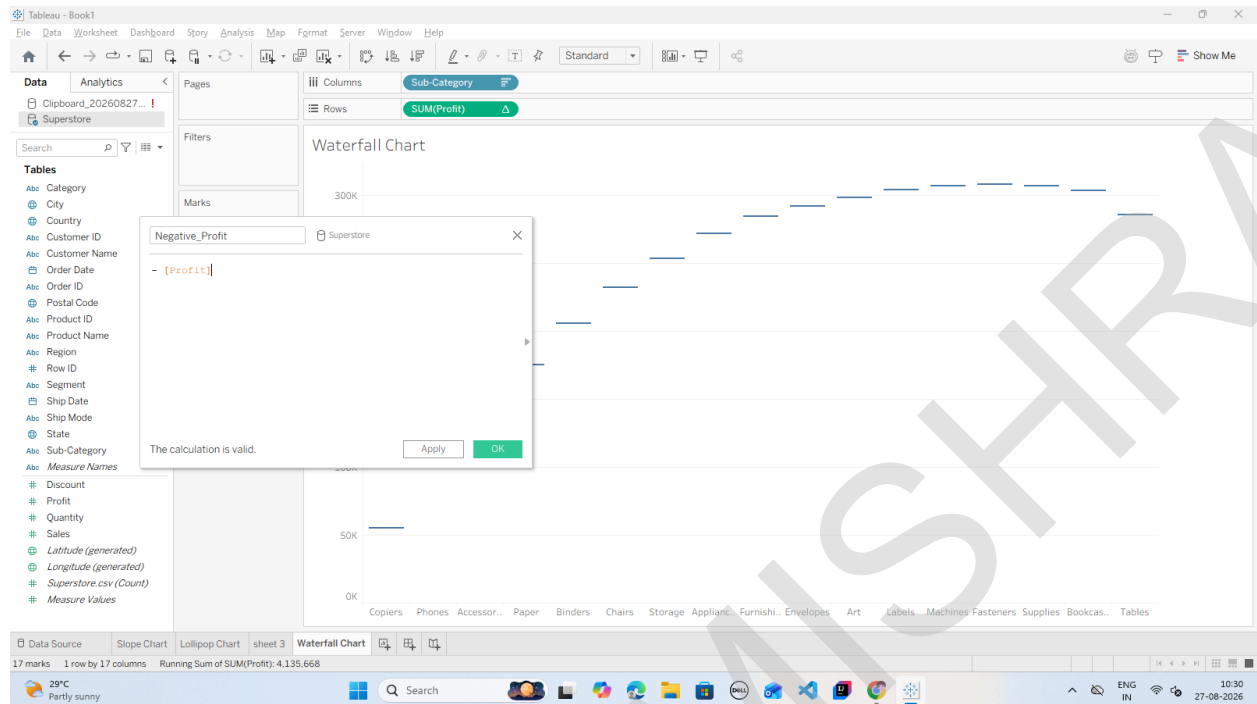
Step 3: Switch Mark Type to Gantt Bar



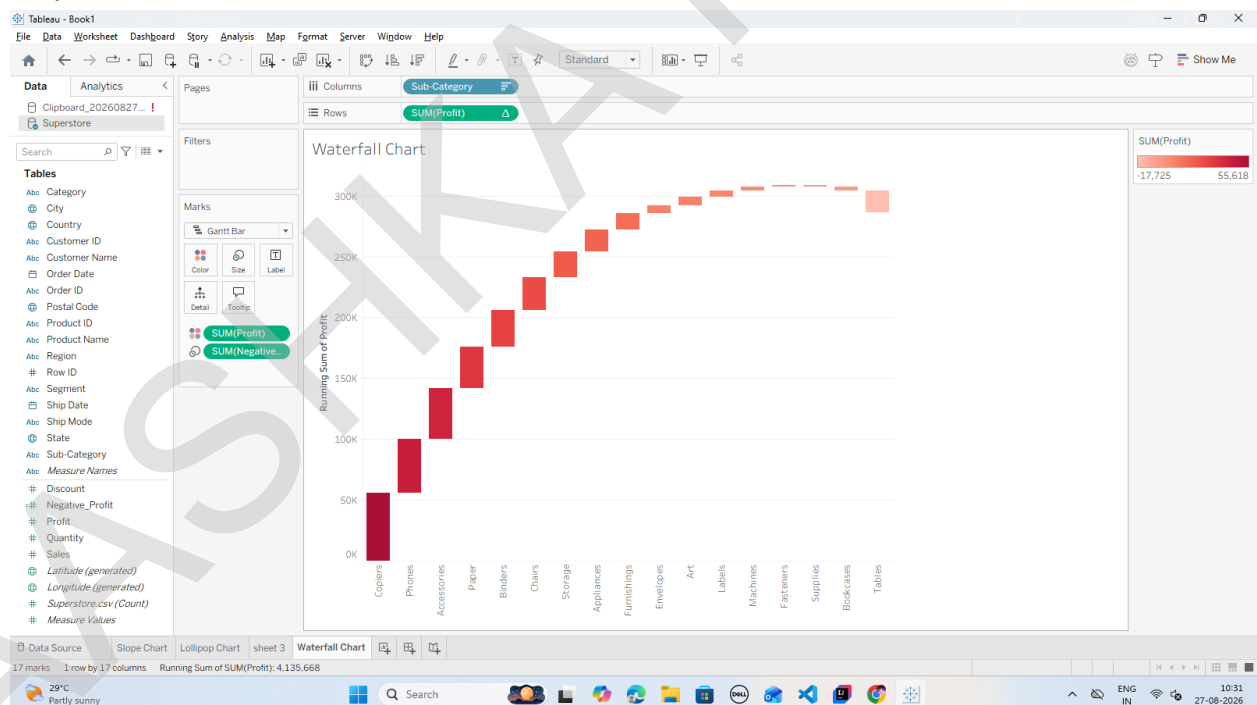
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Step 4: Create the Negative Profit Sizing Calculation



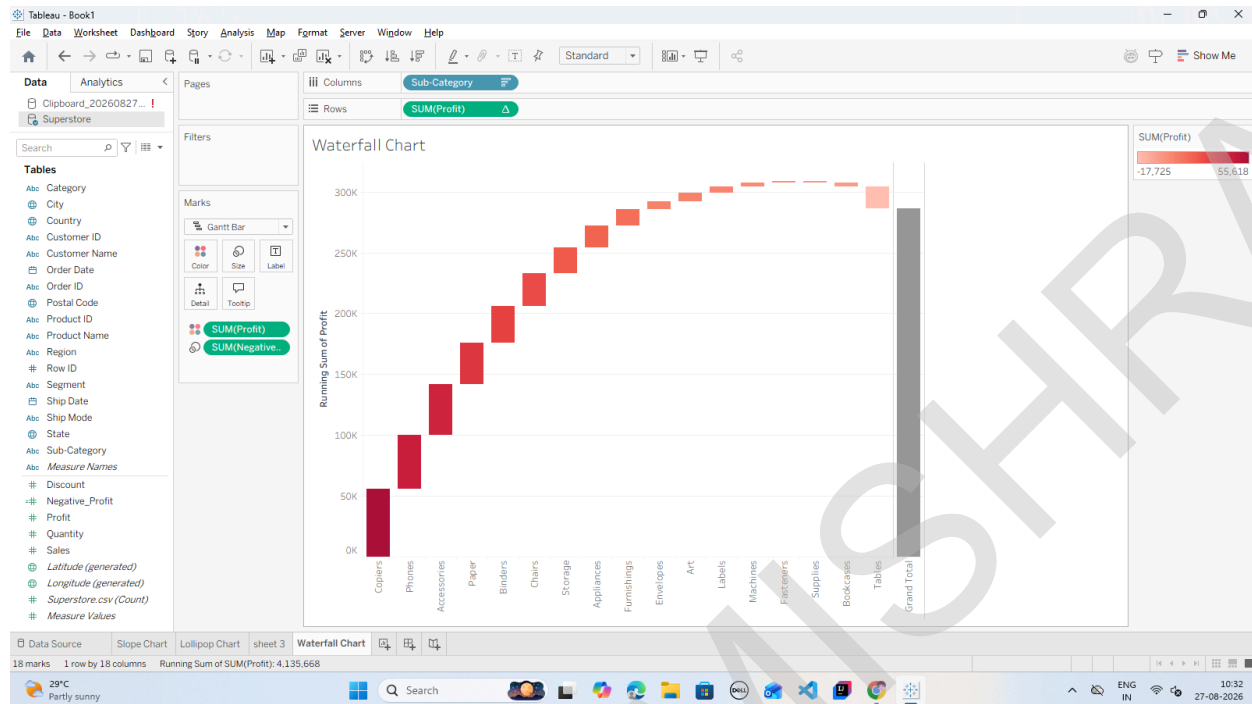
Step 5: Size and Color the Waterfall Blocks



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SUBJECT: Data Visualization with Power BI and Tableau

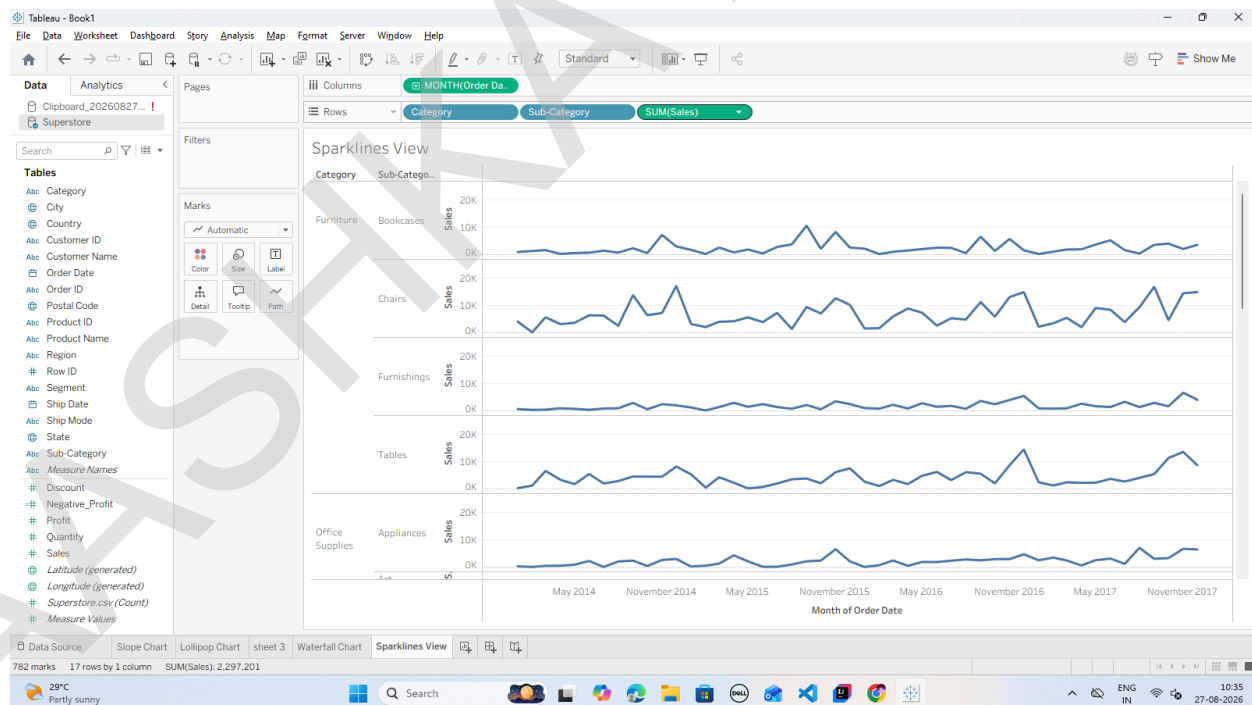
Step 6: Add Grand Totals to Close the Waterfall



C: Build Sparklines for Trend Analysis

Step 1: Set Up the Base Worksheet and Table Dimensions

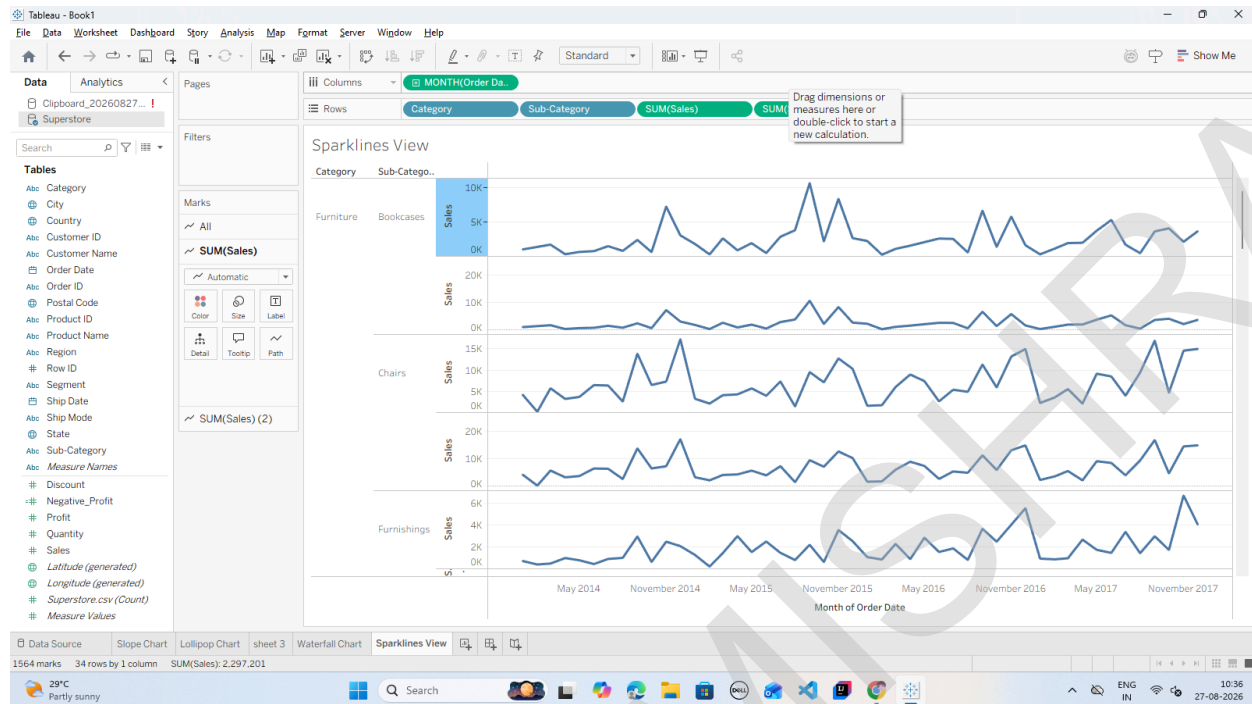
Step 2: Add Continuous Time-Series for Trend Shape



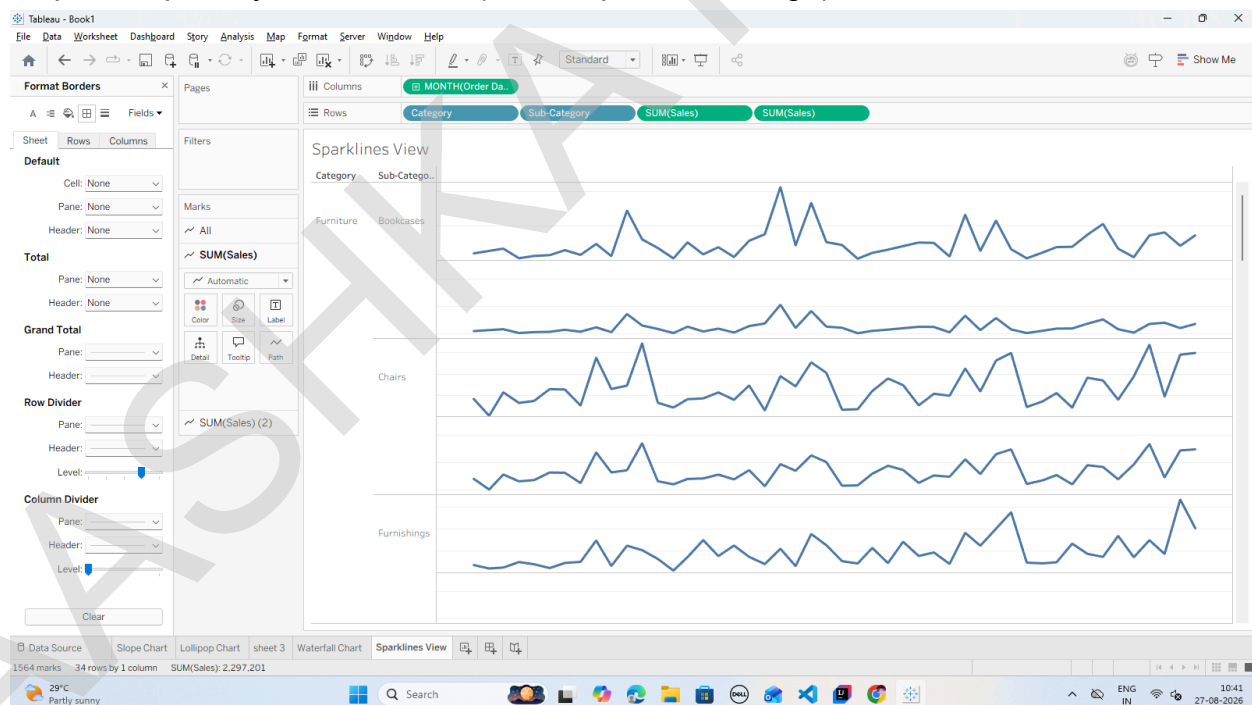
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Step 3: Add Total Sales Metric Pill alongside the Sparkline



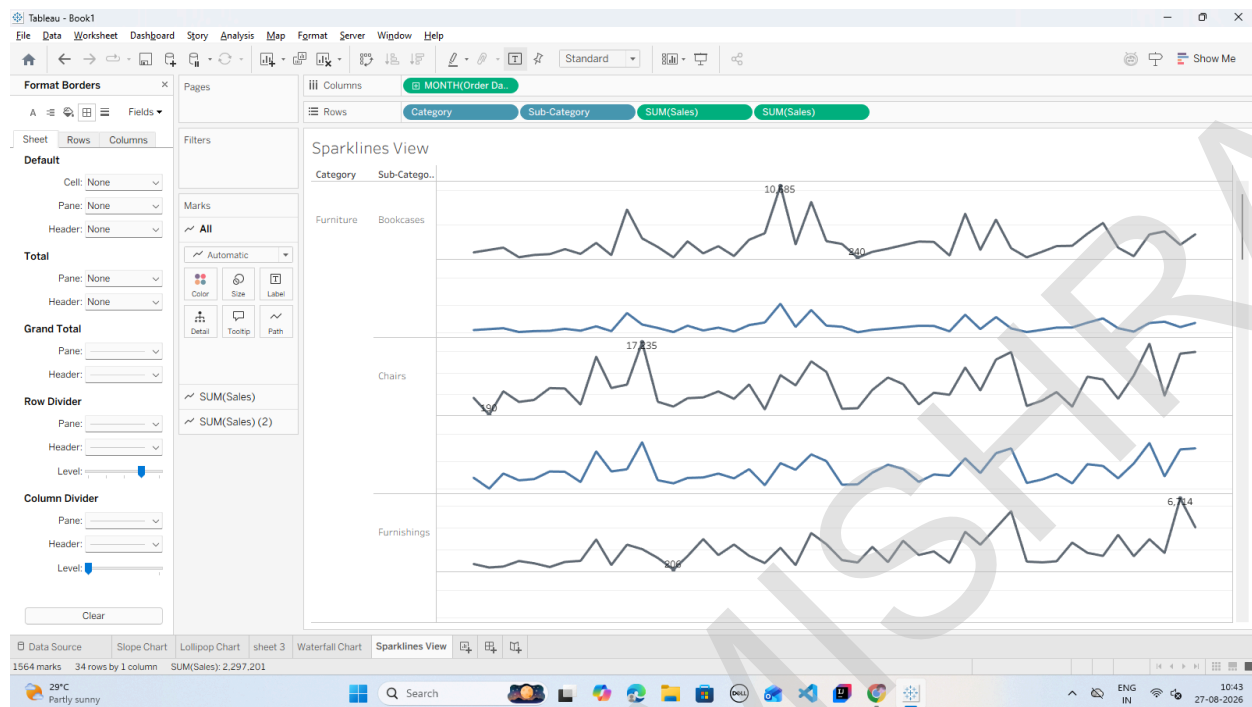
Step 4: Strip Away Visual Clutter (Clean Sparkline Design)



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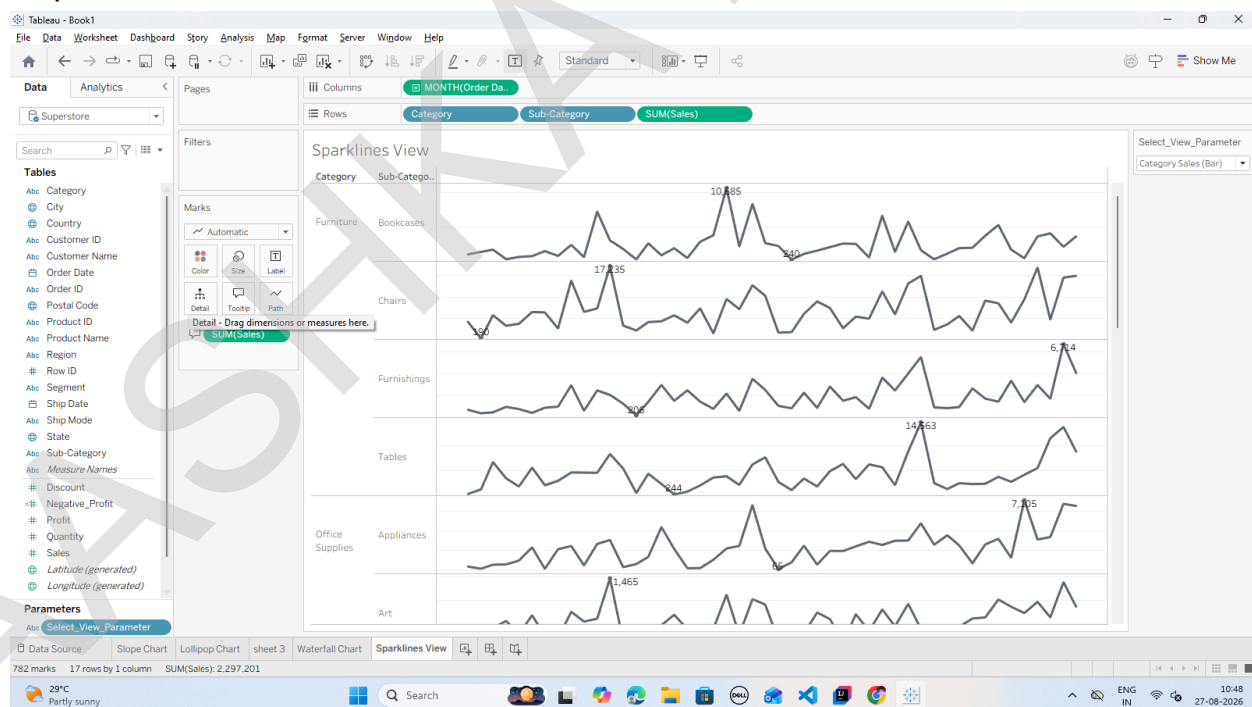
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Step 5: Highlight Minimum and Maximum Milestone Endpoints



D: Implement Sheet Swapping for Dynamic Dashboards

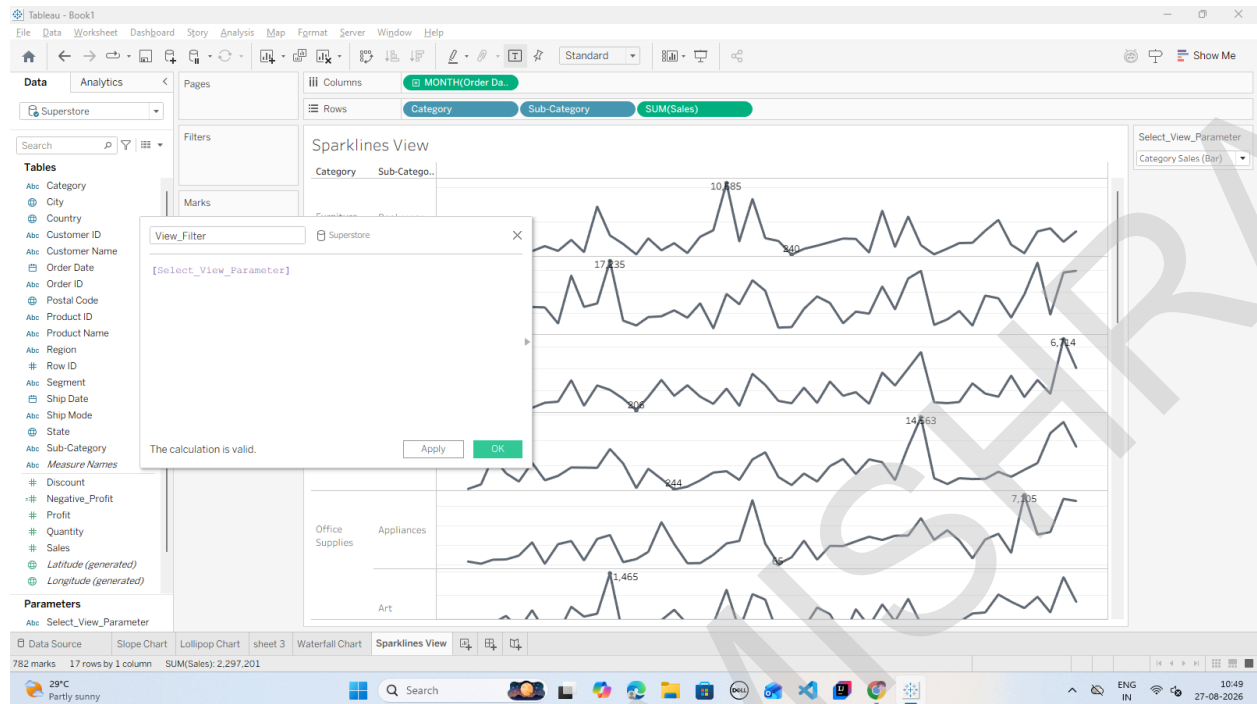
Step 1: Create the Control Parameter



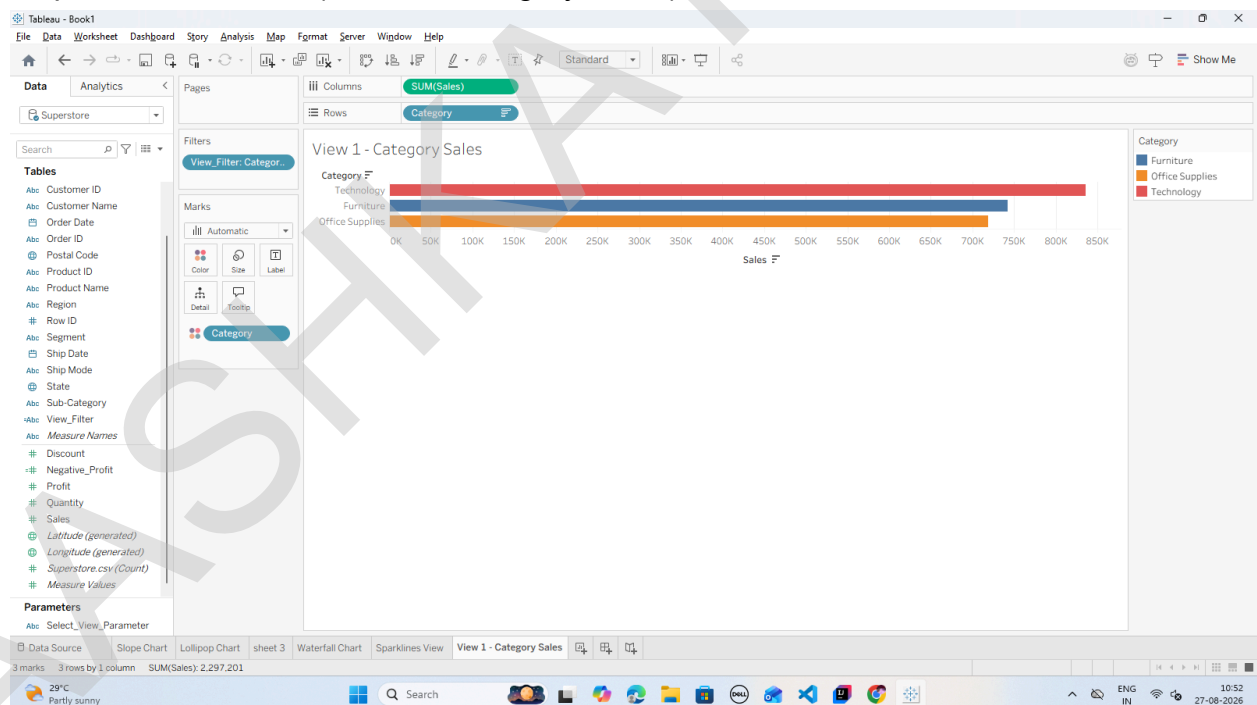
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Step 2: Create the Dynamic Filter Field



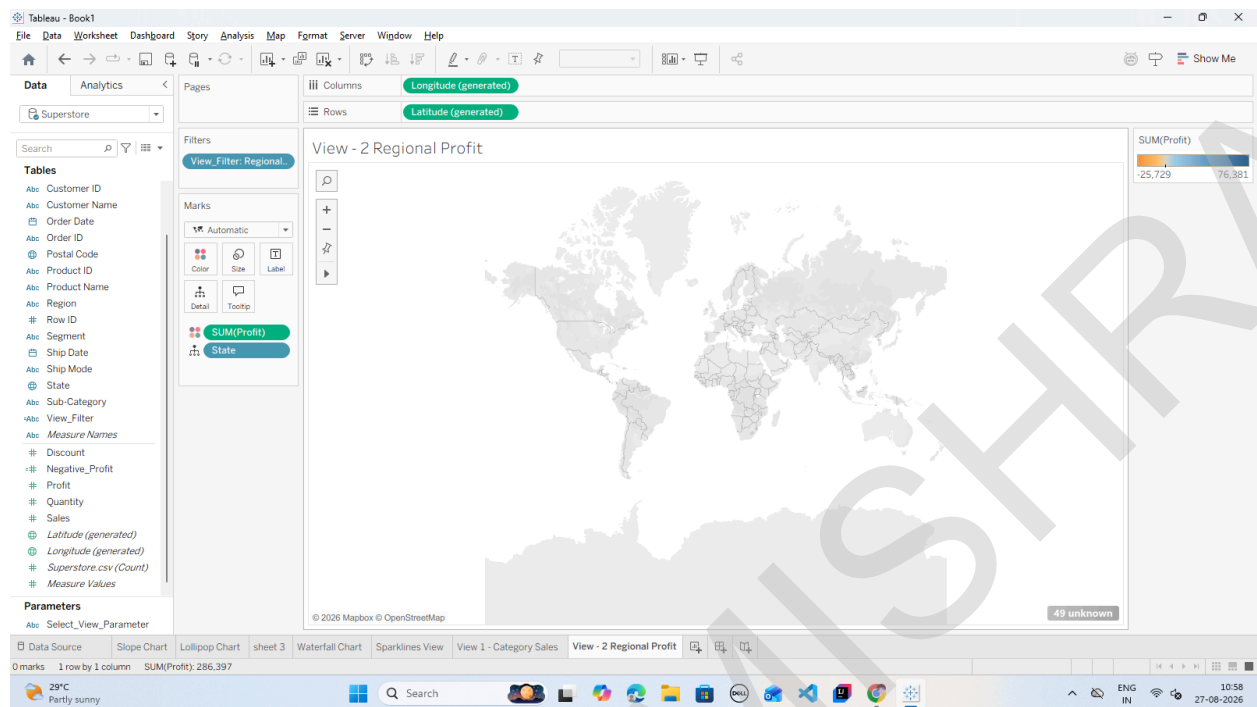
Step 3: Build Sheet 1 (View 1 - Category Sales)



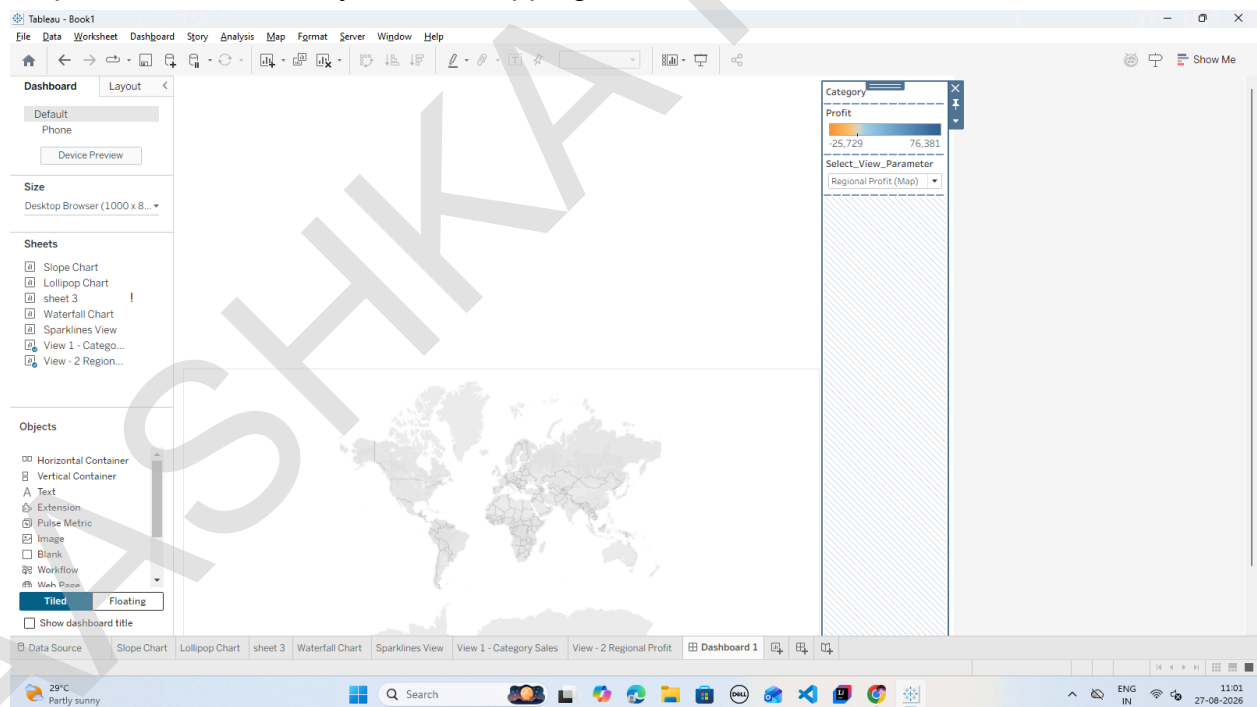
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Step 4: Build Sheet 2 (View 2 - Regional Profit)



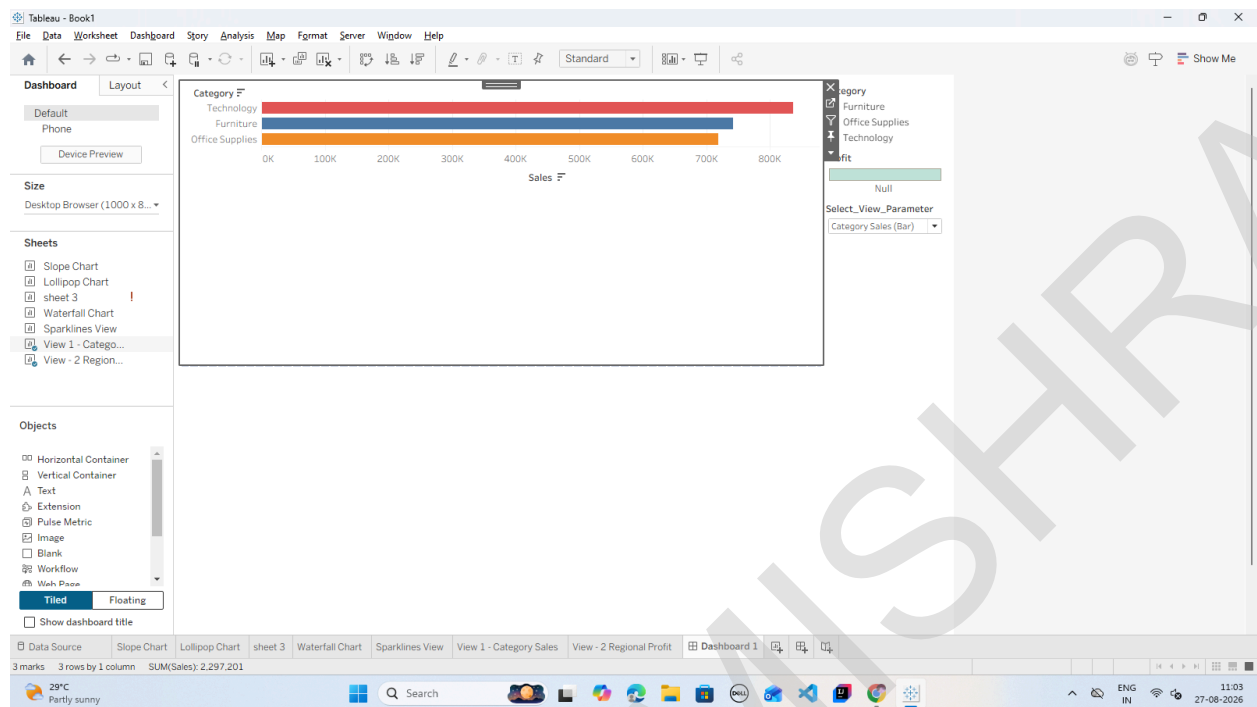
Step 5: Construct the Dynamic Swapping Dashboard



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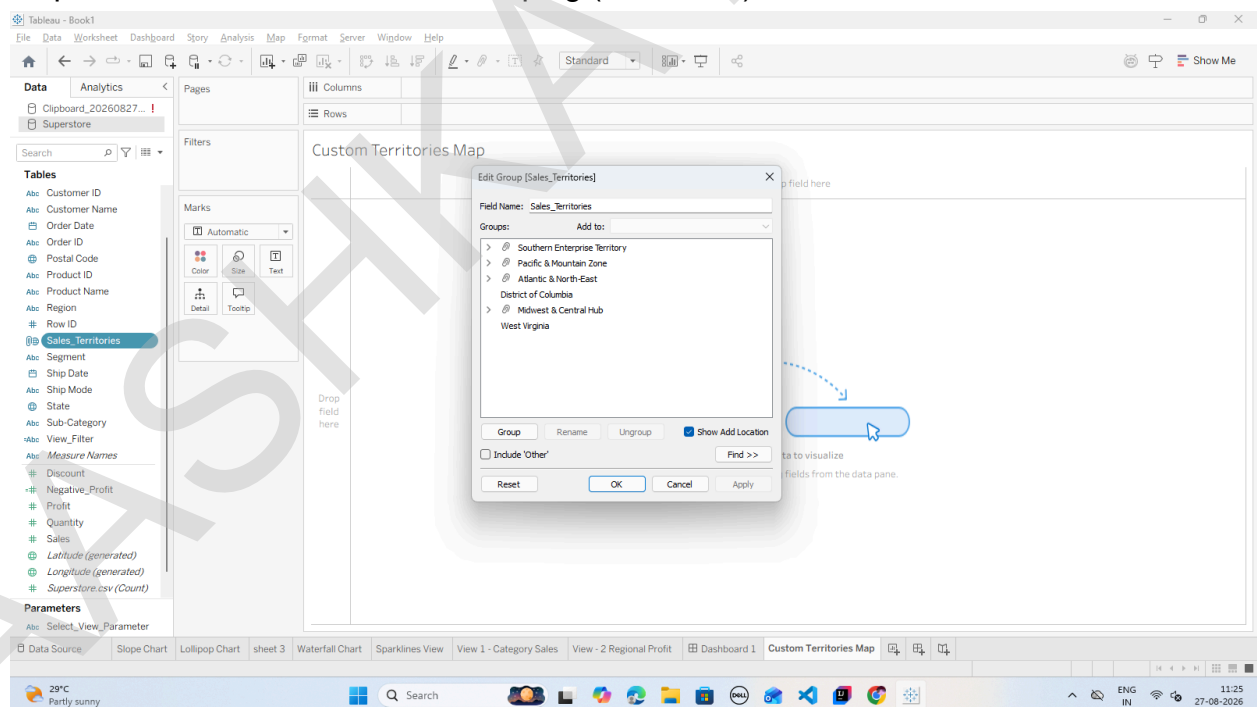
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Step 6: Test the Dynamic Swapping



E: Create Custom Geographic Territories

Step 1: Create a Custom State Grouping (Territories)

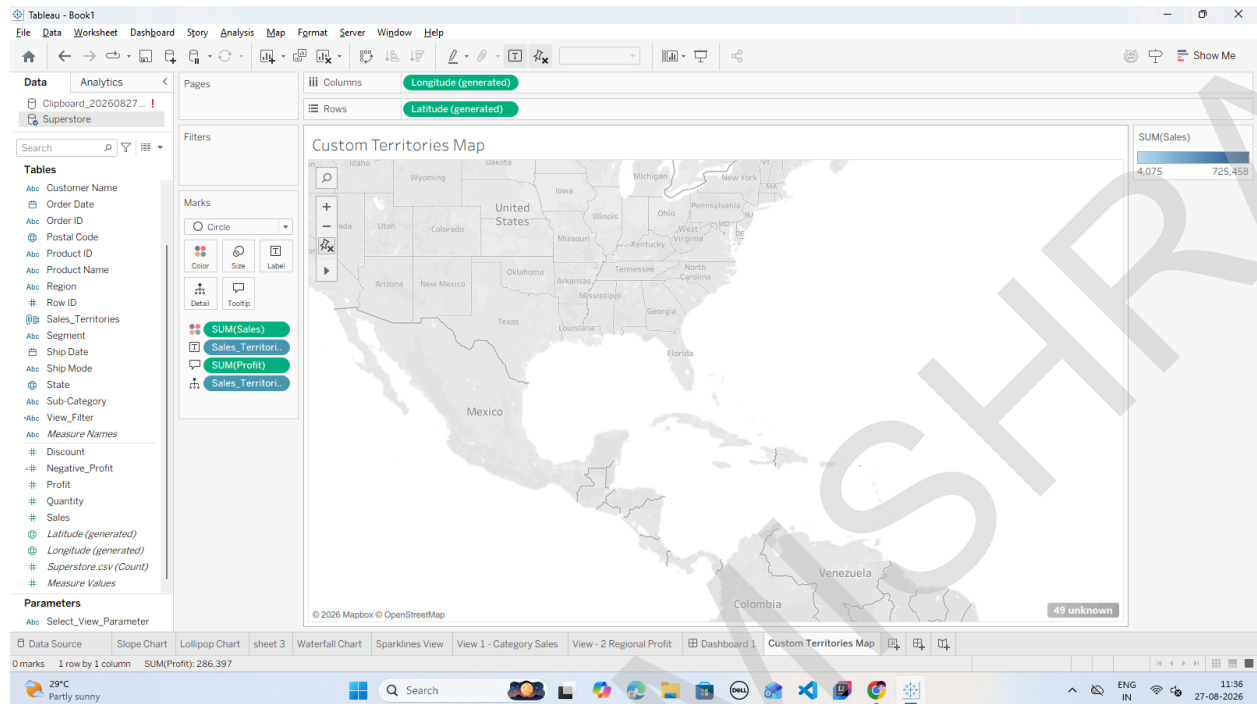


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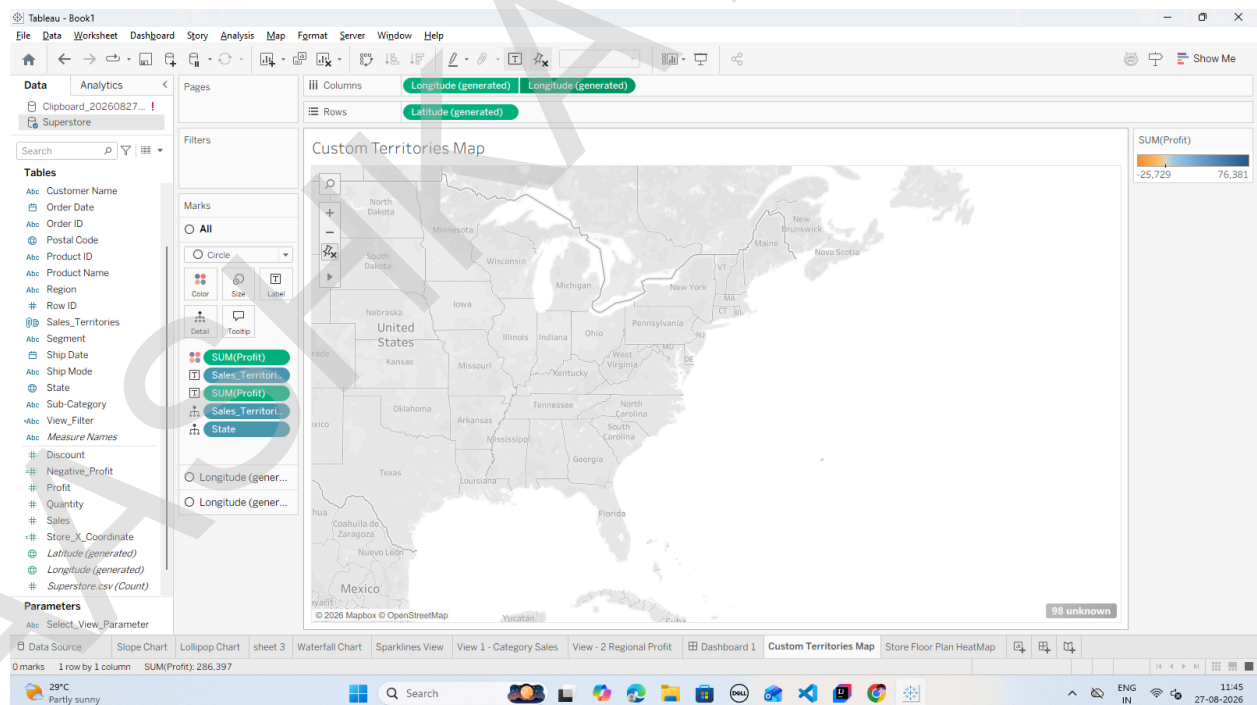
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Step 2: Assign Geographic Role to the Custom Group

Step 3: Build the Consolidated Territory Map



Step 4: Compare State Granularity vs. Custom Territory

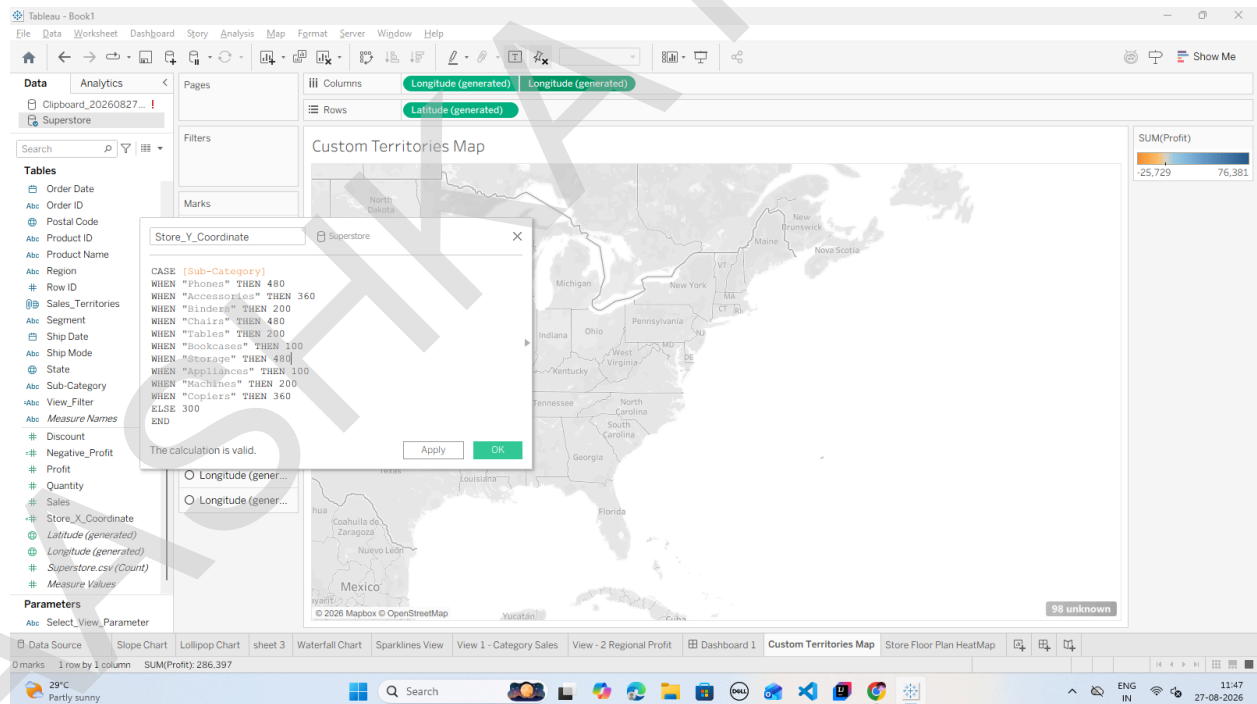
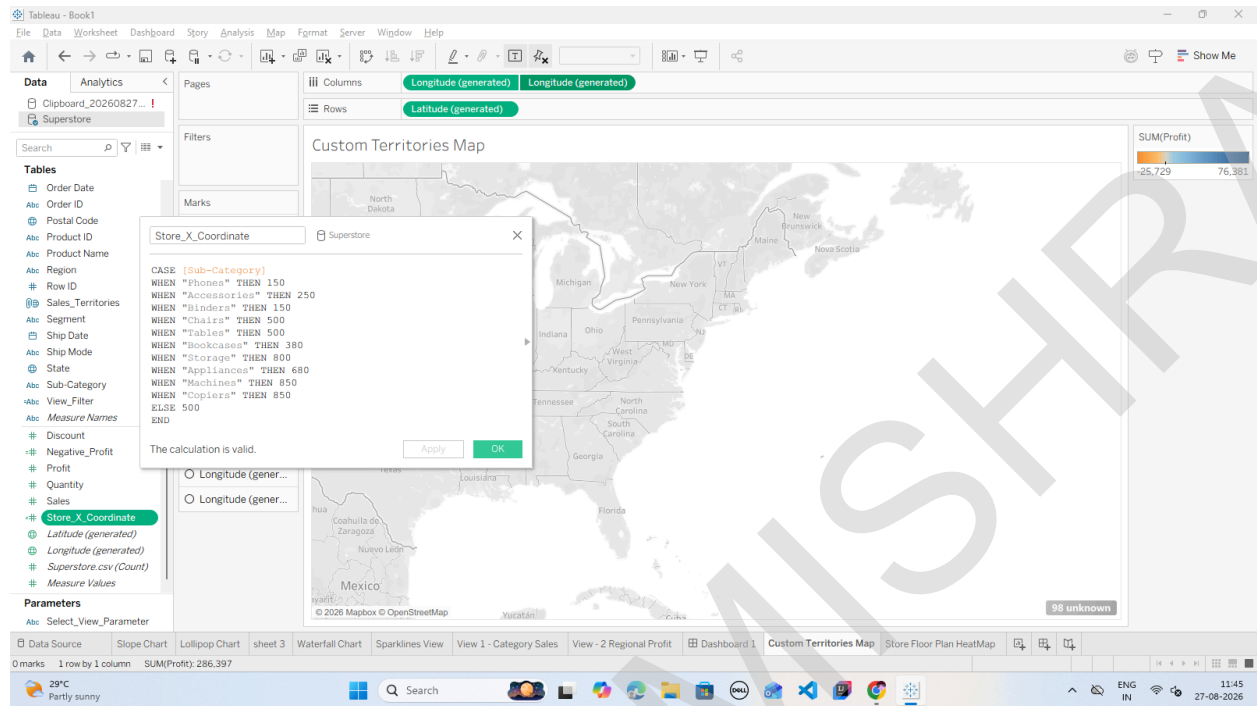


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SUBJECT: Data Visualization with Power BI and Tableau

F: Use Background Images in Dashboards

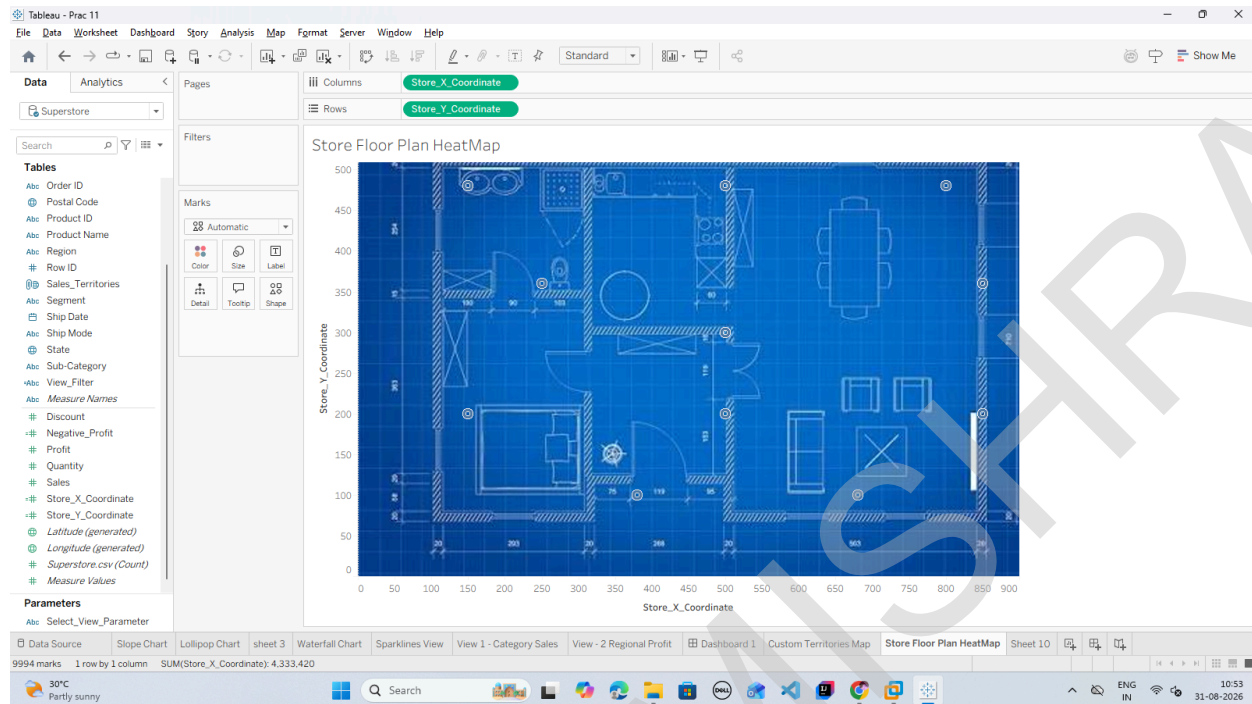
Step 1: Create the Coordinate Calculated Fields



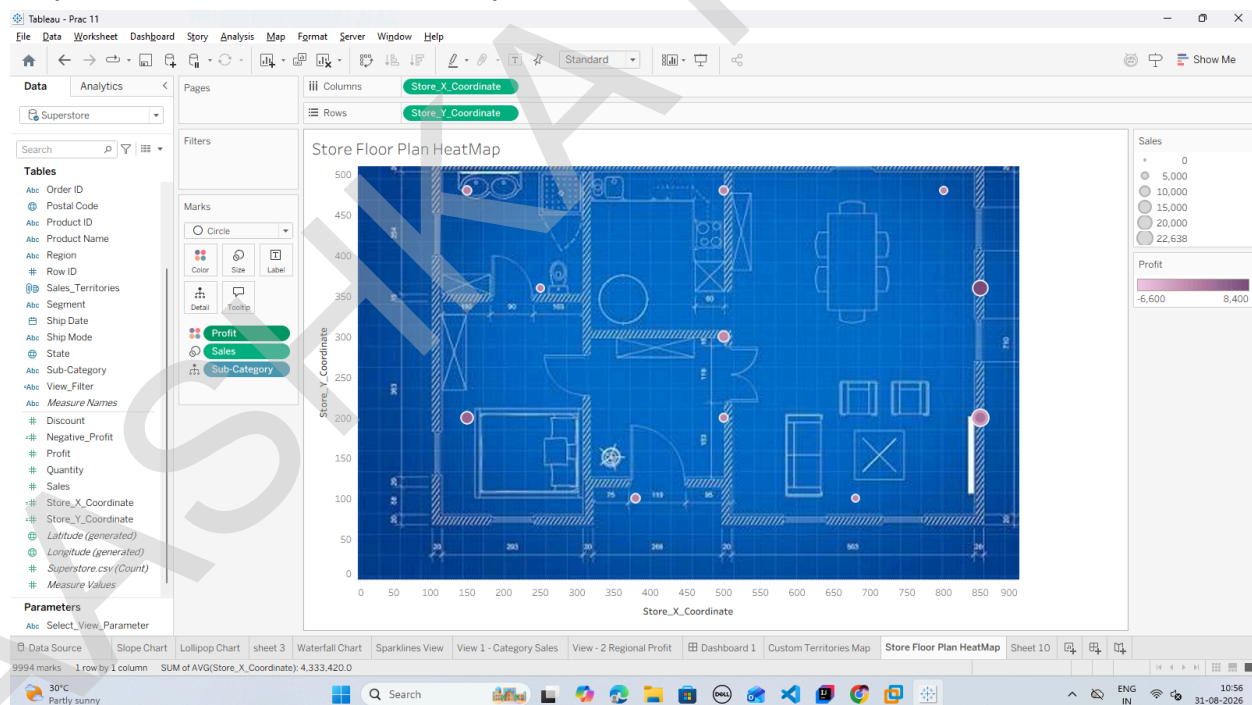
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Step 2: Configure the Custom Background Image in Tableau

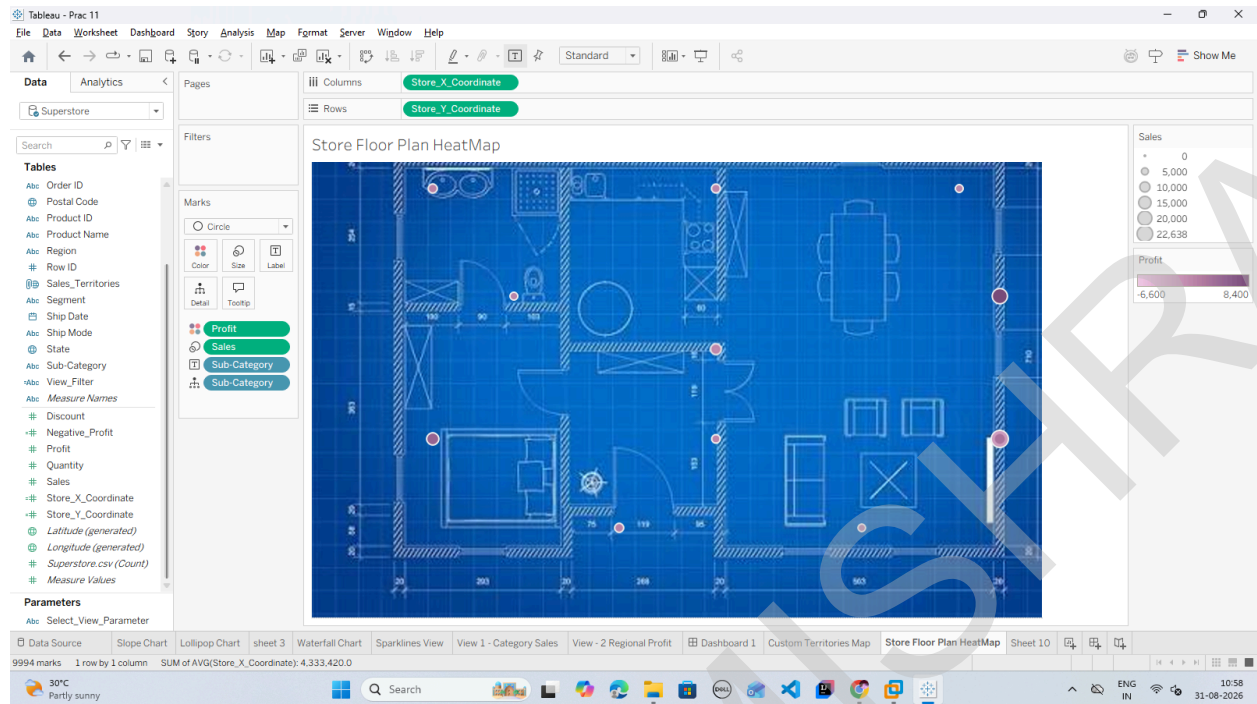


Step 4: Encode Metrics onto the Spatial Zones



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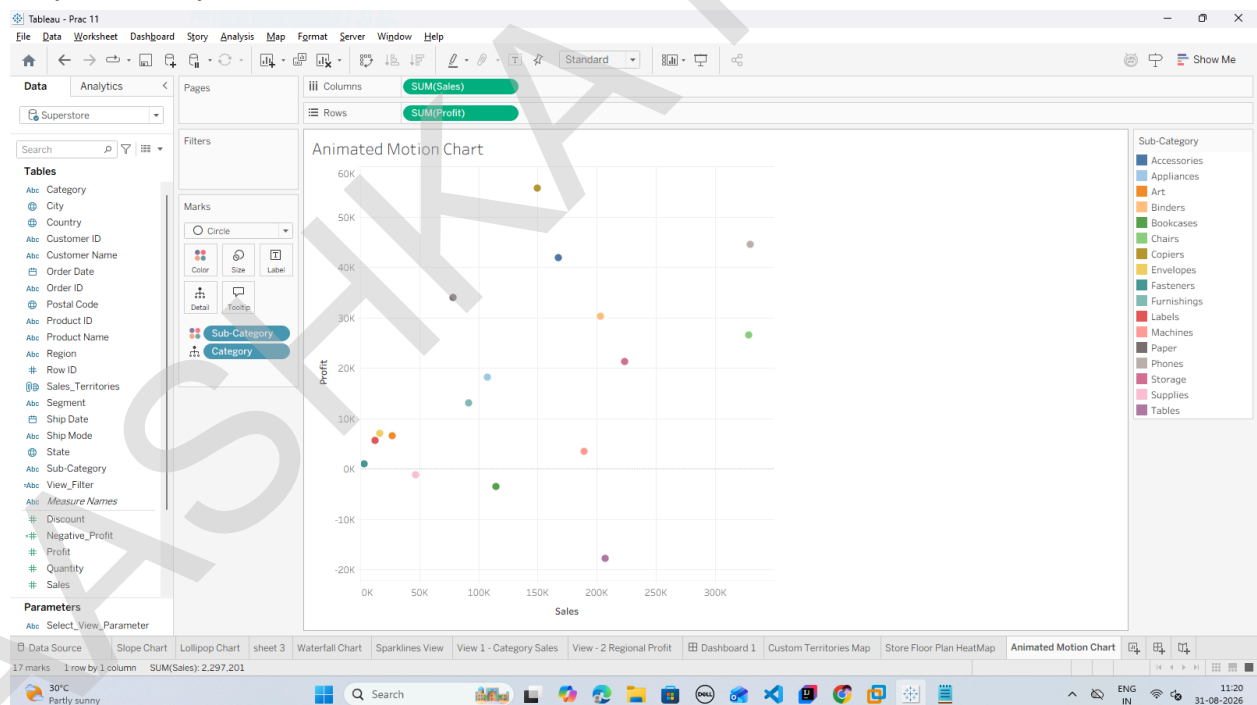
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G: Apply Animation and Transparency Techniques

Task 1: Create an Animated Motion Chart (Pages Shelf)

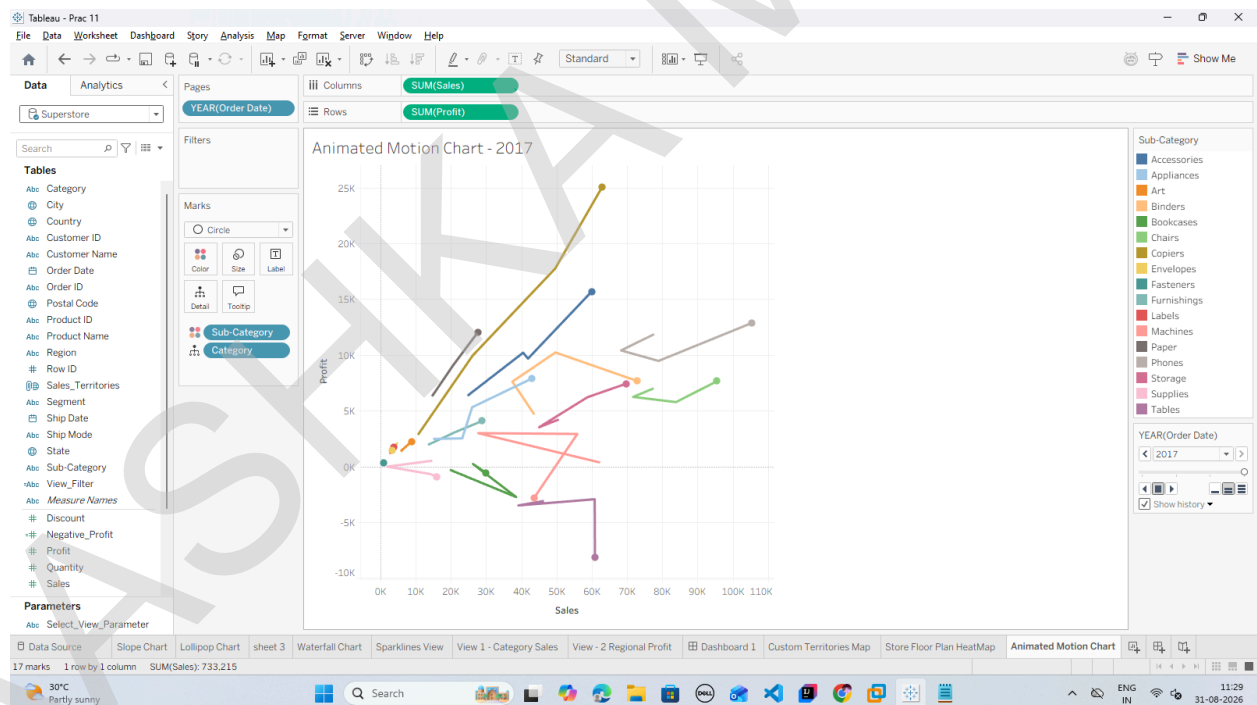
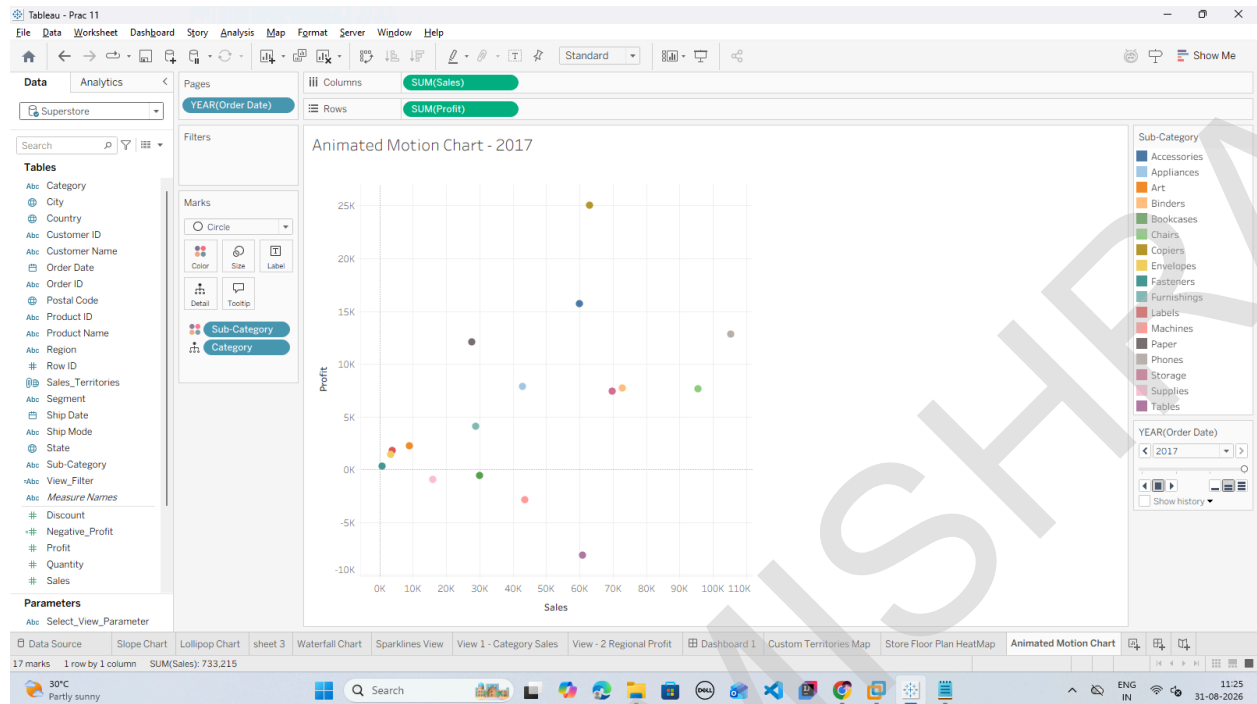
Step 1: Set Up the Scatter Plot Worksheet



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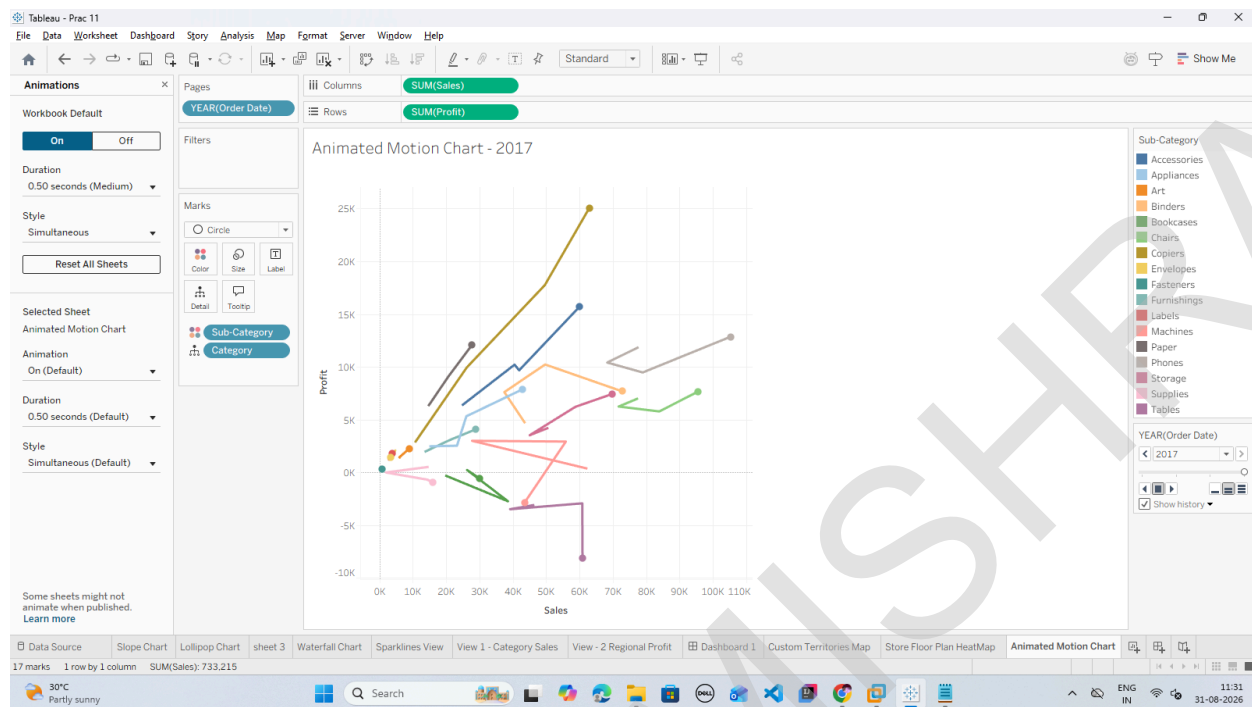
Step 2: Add Time to the Pages Shelf



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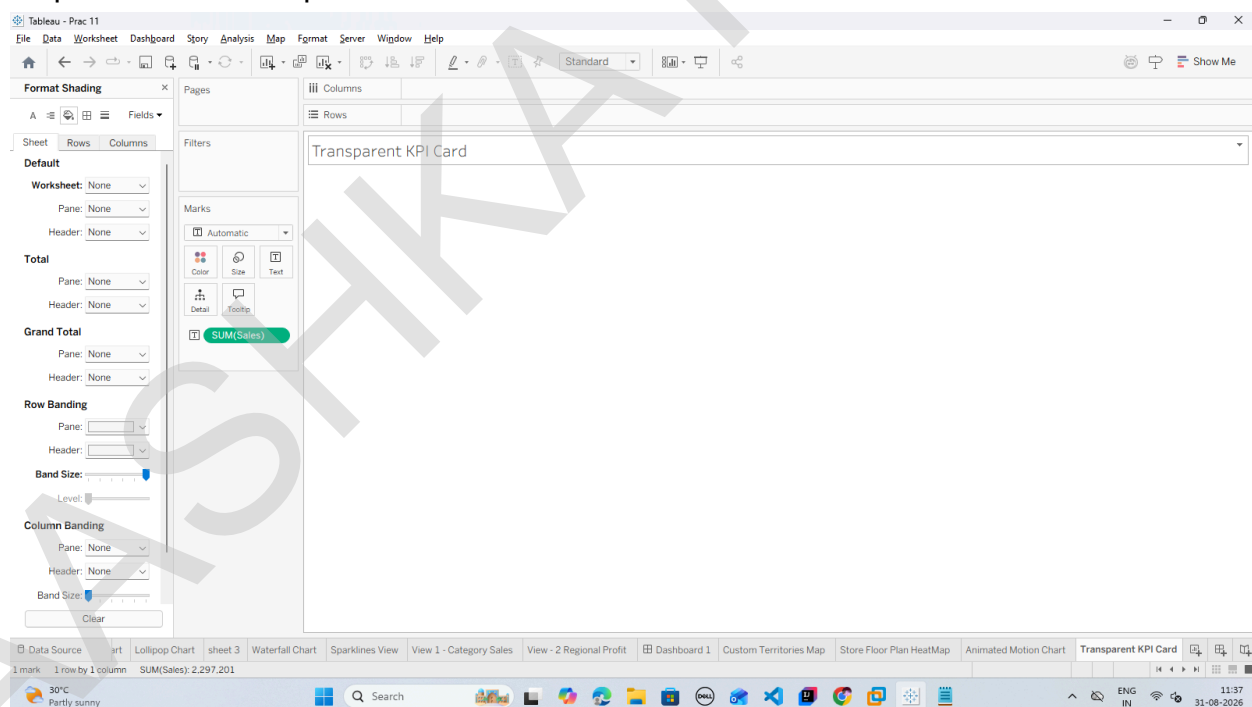
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Step 4: Enable Built-in Tableau Client Animations



Task 2: Apply Sheet and Dashboard Transparency

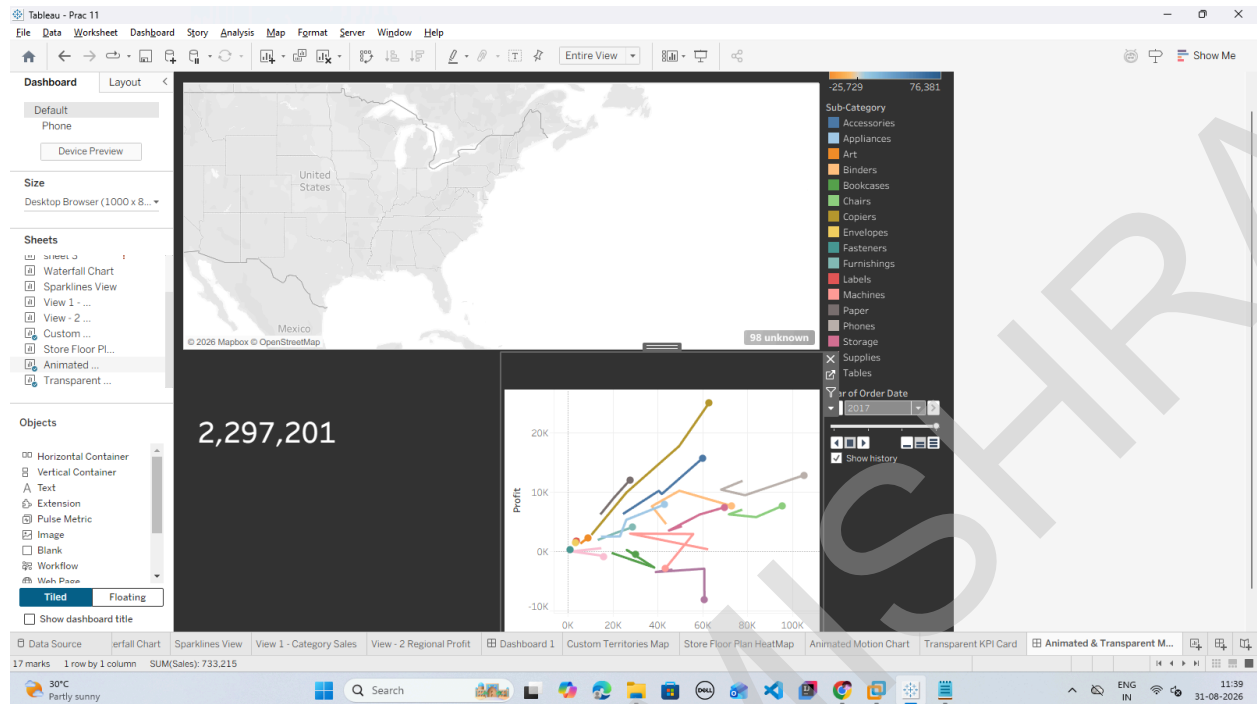
Step 1: Build a Transparent KPI Card Worksheet



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Step 2: Assemble the Transparent Layered Dashboard



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